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Numerical Modeling of Processes with Spatially Fractal Operators.

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Introduction

Mathematical modeling of processes with nonlocal spatial interaction belongs to the relevant directions of modern applied mathematics, since in many media the classical local second-order operator does not make it possible to fully describe the real transport mechanism. In problems of diffusion, filtration, heat conduction, and transport in heterogeneous and multiscale structures, long-range interaction, complex internal geometry, and spatially distributed exchange through internal boundaries play an important role. In such situations, models based on fractional and spatially nonlocal operators turn out to be more accurate; they generalize the classical Laplace operator and make it possible to take into account effects that are not reducible to purely local interaction [4, 9].

Perforated and nanoporous domains with a developed internal structure are relevant for research [5, 7]. In such geometries, the shape of the domain itself influences the transport process. The length of the internal boundaries, self-similarity, and the change in topological complexity during the transition between resolution levels directly affect the character of the solution and the integral characteristics of transport.

The peculiarity of fractional operators on bounded domains lies in the absence of a single canonical realization. For the fractional Laplacian, different definitions correspond to different mechanisms of spatial interaction and different ways of taking the external environment into account [6, 8, 13]. Precisely for this reason, it is meaningful not only to consider the transition itself from a local to a nonlocal model, but also to compare different operator realizations

under consistent boundary conditions and geometric resolution. In nanoporous domains, such a problem becomes even more complicated, since with a change in the geometry level, not only the structure of the domain changes, but also the total intensity of interaction with the internal boundaries. Therefore, a correct numerical study requires such an approach to comparison that makes it possible to separate the effect of the operator itself from trivial geometric scaling.

From the computational point of view, problems with spatially nonlocal operators are significantly more complicated than classical boundary value problems. For local models, the finite element method leads to sparse matrices and relies on a developed theory of approximation [1, 3, 14]. Whereas for nonlocal operators, discretization is, as a rule, associated with dense matrices, double integrals with weak singularity, dependence on the external interaction domain, and increased requirements for the quality of geometric approximation. In perforated and nanoporous domains, this is supplemented by the need to reproduce small geometric elements correctly, to preserve the connectivity of the transport domain, and to avoid numerical artifacts arising from insufficient mesh resolution. Therefore, the construction of effective numerical schemes for such problems is a separate mathematical problem.

This work considers the numerical modeling of diffusion processes in domains with complex internal geometry using local and spatially nonlocal operators of fractional type.

The object of research is stationary and nonstationary transport processes in domains with complex internal geometry, in particular in perforated and nanoporous media, which are described by local and spatially nonlocal operators of fractional type.

The subject of research is mathematical models, variational formulations, numerical schemes, and software implementation of transport problems with local and spatially nonlocal operators of fractional type on domains of complex geometry, including mesh construction, operator discretization, computation of

integral characteristics of the solution, and its visualization.

An additional and no less important aspect is the development of software for visualization, analysis, and modeling of the solution, both stationary and time-dependent.

The aim of the work is to construct and investigate a numerical approach to modeling processes with spatially nonlocal operators on domains of complex geometry, as well as to identify the influence of mesh resolution, the order of the fractional operator, and the type of its realization on the characteristics of transport. To achieve this aim, the corresponding operator formulations are analyzed, their discrete approximations are investigated, computational schemes are implemented for a series of controlled experiments, and the obtained results are compared.

Chapter 1

Problem statement

Let us consider the mathematical formulation of the problems of numerical modeling of diffusion in domains with a complex internal structure for local and spatially nonlocal operators of fractional type.

1.1. Geometric setting

Let

$$\Omega \subset \mathbb{R}^d, \quad d \in \{2, 3\}, \quad (1.1)$$

be a bounded domain with a Lipschitz boundary $\partial\Omega$. In numerical experiments, we focus on the two-dimensional case $d = 2$.

For each fixed $N \geq 0$, we consider a perforated domain

$$D_N = \Omega \setminus \bigcup_{j=1}^{J_N} \overline{\omega_{j,N}}, \quad (1.2)$$

where

$$\omega_{j,N} \Subset \Omega, \quad j = 1, \dots, J_N, \quad (1.3)$$

are open pairwise disjoint subdomains with Lipschitz boundaries, representing internal pores or cavities, and J_N denotes the number of such components for the domain D_N .

The condition $\omega_{j,N} \Subset \Omega$ means that the closure $\overline{\omega_{j,N}}$ is a compact subset of Ω , i.e., each pore is entirely contained within the outer domain and does not touch the boundary $\partial\Omega$.

Thus, the domain D_N is obtained from Ω by removing a finite number of internal subdomains, and it is within this domain that diffusion occurs. Since each $\omega_{j,N}$ has a Lipschitz boundary and $\omega_{j,N} \Subset \Omega$, the domain D_N also has a Lipschitz boundary, allowing for the proper use of traces and integrals over the boundary.

The outer boundary of the domain Ω is denoted by

$$\Gamma_0 := \partial\Omega, \quad (1.4)$$

and the boundary of each internal pore $\omega_{j,N}$ is denoted by

$$\Gamma_{j,N} := \partial\omega_{j,N}, \quad j = 1, \dots, J_N. \quad (1.5)$$

Then

$$\partial D_N = \Gamma_0 \cup \bigcup_{j=1}^{J_N} \Gamma_{j,N}. \quad (1.6)$$

The outer boundary Γ_0 is divided into three pairwise disjoint parts:

$$\Gamma_0 = \Gamma_{\text{in}} \cup \Gamma_{\text{out}} \cup \Gamma_{\text{ins}}, \quad (1.7)$$

where

$$\Gamma_{\text{in}} \cap \Gamma_{\text{out}} = \emptyset, \quad \Gamma_{\text{in}} \cap \Gamma_{\text{ins}} = \emptyset, \quad \Gamma_{\text{out}} \cap \Gamma_{\text{ins}} = \emptyset. \quad (1.8)$$

Here, Γ_{in} is the inlet part of the boundary, Γ_{out} is the outlet part, and Γ_{ins} is the insulated part.

All internal boundaries are combined into the set

$$\Gamma_p(N) := \bigcup_{j=1}^{J_N} \Gamma_{j,N}. \quad (1.9)$$

Thus,

$$\partial D_N = \Gamma_{\text{in}} \cup \Gamma_{\text{out}} \cup \Gamma_{\text{ins}} \cup \Gamma_p(N). \quad (1.10)$$

The outward unit normal to the domain D_N is denoted by $\mathbf{n}$. On Γ_0 , it coincides with the outward normal to Ω , and on $\Gamma_{j,N}$, it is directed towards the pores.

For further analysis, we introduce two geometric characteristics of the domain D_N .

The first one is porosity:

$$\phi_N := \frac{|\Omega \setminus D_N|}{|\Omega|} = \frac{1}{|\Omega|} \sum_{j=1}^{J_N} |\omega_{j,N}|. \quad (1.11)$$

Here, $|\cdot|$ denotes the Lebesgue measure in $\mathbb{R}^d$, i.e., the area for $d = 2$ or the volume for $d = 3$. The quantity ϕ_N indicates the fraction of the outer domain Ω occupied by all internal pores.

The second characteristic is the total measure of the internal boundaries:

$$\Sigma_N := \sum_{j=1}^{J_N} \mathcal{H}^{d-1}(\Gamma_{j,N}), \quad (1.12)$$

where $\mathcal{H}^{d-1}$ is the $(d-1)$ -dimensional Hausdorff measure. In the two-dimensional case, $\mathcal{H}^1$ is the length, so Σ_N equals the total length of all internal boundaries. In the three-dimensional case, $\mathcal{H}^2$ is the surface area, and then Σ_N equals the total area of all internal surfaces.

The quantity Σ_N gives the total $(d-1)$ -dimensional measure of the internal boundaries, that is, a natural geometric characteristic of the contact set between the transport region and the internal pores [10]. It is used for normalizing the integral characteristics of the solution and for consistently defining the exchange through the internal boundaries.

1.2. Local Stationary Diffusion Model

We start with the classical local stationary diffusion model in the perforated domain D_N . The unknown function

$$u_N : D_N \rightarrow \mathbb{R} \quad (1.13)$$

describes the scalar state of the medium at each point in the domain. The function u_N can be interpreted as the concentration of a substance, temperature, pressure, chemical potential, or another scalar characteristic that determines the direction and intensity of the diffusion flux.

In the general case, the local stationary model in the domain D_N is given by the equation

$$-\nabla \cdot (a_N(x) \nabla u_N(x)) + c_N(x) u_N(x) = f_N(x), \quad x \in D_N. \quad (1.14)$$

Function

$$a_N : D_N \rightarrow \mathbb{R} \quad (1.15)$$

is interpreted as the diffusion coefficient. It is assumed that $a_N \in L^\infty(D_N)$ and

$$a_N(x) \geq a_* > 0 \quad \text{for almost all } x \in D_N, \quad (1.16)$$

where the constant a_* does not depend on x . Condition (1.16) means that the medium does not lose its ability to diffuse at any point in the domain, i.e., the equation remains uniformly elliptic.

Function

$$c_N : D_N \rightarrow \mathbb{R}, \quad c_N(x) \geq 0, \quad (1.17)$$

is the absorption coefficient; it is assumed that $c_N \in L^\infty(D_N)$.

Function

$$f_N : D_N \rightarrow \mathbb{R} \quad (1.18)$$

describes the volumetric source; it is assumed that $f_N \in L^2(D_N)$.

Next, we formulate the boundary conditions. On the inlet and outlet parts of the external boundary, Dirichlet conditions are imposed:

$$u_N = U_{\text{in}} \quad \text{on } \Gamma_{\text{in}}, \quad (1.19)$$

$$u_N = U_{\text{out}} \quad \text{on } \Gamma_{\text{out}}, \quad (1.20)$$

where the numbers

$$U_{\text{in}}, U_{\text{out}} \in \mathbb{R} \quad (1.21)$$

are the specified potential levels on the corresponding parts of the boundary. It is the difference between these values that creates the macroscopic load on the domain. We denote it by

$$\Delta U := U_{\text{in}} - U_{\text{out}}. \quad (1.22)$$

The quantity ΔU will henceforth play the role of the external potential difference that drives diffusion through the domain D_N .

On the isolated part of the external boundary, a Neumann condition is imposed

$$a_N \nabla u_N \cdot \mathbf{n} = 0 \quad \text{on } \Gamma_{\text{ins}}, \quad (1.23)$$

where $\mathbf{n}$ is the outward unit normal to the domain D_N . The scalar product $\nabla u_N \cdot \mathbf{n}$ represents the normal derivative of the function u_N along the boundary, and therefore the condition (1.23) means that there is no flux through the isolated part of the boundary.

On each internal boundary $\Gamma_{j,N}$, corresponding to the boundary of the j -th pore or internal inclusion, a Robin condition is imposed

$$a_N \nabla u_N \cdot \mathbf{n} + \kappa_{j,N} u_N = \kappa_{j,N} u_{b,j,N}, \quad \text{on } \Gamma_{j,N}, \quad j = 1, \dots, J_N. \quad (1.24)$$

Here

$$\kappa_{j,N} \geq 0 \quad (1.25)$$

is the exchange coefficient on the boundary $\Gamma_{j,N}$, and

$$u_{b,j,N} \in \mathbb{R} \quad (1.26)$$

is the specified level of the external environment or reservoir with which exchange occurs through this boundary. The larger the coefficient $\kappa_{j,N}$, the more intensely the value of u_N on the boundary $\Gamma_{j,N}$ is attracted to the level $u_{b,j,N}$.

If we move from the representation on individual components to the representation on the entire pore boundary

$$\Gamma_p(N) = \bigcup_{j=1}^{J_N} \Gamma_{j,N}, \quad (1.27)$$

then the condition (1.24) can be presented in a more compact form:

$$a_N \nabla u_N \cdot \mathbf{n} + \kappa_N u_N = \kappa_N u_{b,N} \quad \text{on } \Gamma_p(N), \quad (1.28)$$

where the functions

$$\kappa_N : \Gamma_p(N) \rightarrow \mathbb{R}, \quad u_{b,N} : \Gamma_p(N) \rightarrow \mathbb{R} \quad (1.29)$$

are piecewise constant on the components $\Gamma_{j,N}$, i.e., on each of them they take constant values

$$\kappa_N|_{\Gamma_{j,N}} = \kappa_{j,N}, \quad u_{b,N}|_{\Gamma_{j,N}} = u_{b,j,N}.$$

To move to the weak formulation, it is necessary to take into account that the boundary values for functions from the Sobolev space are understood in the sense of traces. Let $H^1(D_N)$ denote the standard space of functions whose first generalized derivatives are square-integrable in the domain D_N .

The set of admissible functions, taking into account the inhomogeneous Dirichlet conditions, is defined as

$$V_N := \{v \in H^1(D_N) : v = U_{\text{in}} \text{ on } \Gamma_{\text{in}}, \quad v = U_{\text{out}} \text{ on } \Gamma_{\text{out}}\}. \quad (1.30)$$

Here, the equalities on the boundary should be understood in the sense of the trace of the function v on the corresponding parts of the boundary.

The corresponding space of test functions is

$$V_{N,0} := \{v \in H^1(D_N) : v = 0 \text{ on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}\}. \quad (1.31)$$

Functions from the space $V_{N,0}$ vanish on the parts of the boundary where Dirichlet conditions are imposed for the sought solution.

Now we can write the weak formulation of the local problem. We need to find a function

$$u_N \in V_N, \quad (1.32)$$

such that for every test function

$$v \in V_{N,0} \quad (1.33)$$

the following identity holds

$$\begin{aligned} \int_{D_N} a_N \nabla u_N \cdot \nabla v \, dx + \int_{D_N} c_N u_N v \, dx \\ + \int_{\Gamma_p(N)} \kappa_N u_N v \, d\sigma \\ = \int_{D_N} f_N v \, dx + \int_{\Gamma_p(N)} \kappa_N u_{b,N} v \, d\sigma. \end{aligned} \quad (1.34)$$

In the identity (1.34), the first integral on the left corresponds to diffusion in the volume, the second to absorption, and the third to exchange through the pore boundary. The right-hand side describes the volumetric source and the influence of the external environment on the pore boundaries.

1.3. Spatially Nonlocal Operators of Fractional Type

To take into account nonlocal spatial interaction, instead of a local second-order operator, we consider operators of fractional type. On a bounded domain, the fractional analogue of the Laplace operator is defined not in a unique way, therefore in what follows three formulations are used: integral, regional, and spectral [4, 8, 9].

Let

$$s \in (0, 1) \quad (1.35)$$

be the parameter of fractionality. In the space $\mathbb{R}^d$, $d \in \{2, 3\}$, let us introduce the kernel

$$K_s(x, y) = \frac{C_{d,s}}{|x - y|^{d+2s}}, \quad x \neq y, \quad (1.36)$$

where $C_{d,s} > 0$ is a normalizing constant.

1.3.1. Integral Realization

Let

$$\Lambda_N \subset \mathbb{R}^d, \quad \overline{D_N} \subset \Lambda_N, \quad (1.37)$$

where Λ_N is an extended domain containing D_N and the outer interaction zone. For $u, v \in H^s(\Lambda_N)$, we define the bilinear form

$$a_{s,N}^{\text{int}}(u, v) = \frac{1}{2} \iint_{\Lambda_N \times \Lambda_N} (u(x) - u(y))(v(x) - v(y)) K_s(x, y) dx dy. \quad (1.38)$$

Exchange with the external environment is prescribed on the set

$$E_N^{\text{int}} \subset \Lambda_N \setminus D_N \quad (1.39)$$

with the help of the functions

$$\kappa_N^{\text{int}} : E_N^{\text{int}} \rightarrow [0, +\infty), \quad u_{b,N}^{\text{int}} : E_N^{\text{int}} \rightarrow \mathbb{R}. \quad (1.40)$$

Accordingly,

$$r_{s,N}^{\text{int}}(u, v) = \int_{E_N^{\text{int}}} \kappa_N^{\text{int}}(x) u(x) v(x) dx, \quad (1.41)$$

$$\ell_{s,N}^{\text{int}}(v) = \int_{D_N} f_N(x) v(x) dx + \int_{E_N^{\text{int}}} \kappa_N^{\text{int}}(x) u_{b,N}^{\text{int}}(x) v(x) dx. \quad (1.42)$$

One seeks a function

$$u_{N,s}^{\text{int}} \in V_{N,s}^{\text{int}} \subset H^s(\Lambda_N), \quad (1.43)$$

which satisfies

$$a_{s,N}^{\text{int}}(u_{N,s}^{\text{int}}, v) + r_{s,N}^{\text{int}}(u_{N,s}^{\text{int}}, v) = \ell_{s,N}^{\text{int}}(v) \quad \forall v \in V_{N,s,0}^{\text{int}}. \quad (1.44)$$

For

$$s > \frac{1}{2} \quad (1.45)$$

one can set

$$V_{N,s}^{\text{int}} = \{v \in H^s(\Lambda_N) : v = U_{\text{in}} \text{ on } \Gamma_{\text{in}}, \quad v = U_{\text{out}} \text{ on } \Gamma_{\text{out}}\}, \quad (1.46)$$

$$V_{N,s,0}^{\text{int}} = \{v \in H^s(\Lambda_N) : v = 0 \text{ on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}\}. \quad (1.47)$$

For

$$0 < s \leq \frac{1}{2} \quad (1.48)$$

the Dirichlet conditions are not prescribed directly through the trace on the boundary; in this case they are taken into account with the help of an auxiliary function or already at the level of the discrete formulation. The unknown function in the integral model is defined on the whole domain Λ_N .

1.3.2. Regional Realization

In the regional formulation, interaction takes place only inside D_N . For $u, v \in H^s(D_N)$, we have

$$a_{s,N}^{\text{reg}}(u, v) = \frac{1}{2} \iint_{D_N \times D_N} (u(x) - u(y))(v(x) - v(y)) K_s(x, y) dx dy. \quad (1.49)$$

If

$$s > \frac{1}{2}, \quad (1.50)$$

then exchange through the pore boundary is prescribed by the boundary term

$$r_{s,N}^{\text{reg}}(u, v) = \int_{\Gamma_{\text{p}}(N)} \kappa_N^{\text{reg}} u v d\sigma, \quad (1.51)$$

$$\ell_{s,N}^{\text{reg}}(v) = \int_{D_N} f_N v \, dx + \int_{\Gamma_p(N)} \kappa_N^{\text{reg}} u_{b,N}^{\text{reg}} v \, d\sigma. \quad (1.52)$$

If

$$s \leq \frac{1}{2}, \quad (1.53)$$

then instead of the boundary term an inner layer is introduced

$$(D_N)_\delta = \{x \in D_N : \text{dist}(x, \partial D_N) < \delta\}, \quad \delta > 0, \quad (1.54)$$

and then

$$r_{s,N}^{\text{lay}}(u, v) = \int_{(D_N)_\delta} \kappa_{N,\delta}^{\text{lay}}(x) u(x) v(x) \, dx, \quad (1.55)$$

$$\ell_{s,N}^{\text{lay}}(v) = \int_{D_N} f_N(x) v(x) \, dx + \int_{(D_N)_\delta} \kappa_{N,\delta}^{\text{lay}}(x) u_{b,N,\delta}^{\text{lay}}(x) v(x) \, dx. \quad (1.56)$$

One seeks

$$u_{N,s}^{\text{reg}} \in V_{N,s}^{\text{reg}} \subset H^s(D_N), \quad (1.57)$$

such that

$$a_{s,N}^{\text{reg}}(u_{N,s}^{\text{reg}}, v) + r_{s,N}^{\text{reg},*}(u_{N,s}^{\text{reg}}, v) = \ell_{s,N}^{\text{reg},*}(v) \quad \forall v \in V_{N,s,0}^{\text{reg}}, \quad (1.58)$$

where the symbol $*$ denotes the boundary or layer variant.

For

$$s > \frac{1}{2} \quad (1.59)$$

the spaces can be written as

$$V_{N,s}^{\text{reg}} = \{v \in H^s(D_N) : v = U_{\text{in}} \text{ on } \Gamma_{\text{in}}, \quad v = U_{\text{out}} \text{ on } \Gamma_{\text{out}}\}, \quad (1.60)$$

$$V_{N,s,0}^{\text{reg}} = \{v \in H^s(D_N) : v = 0 \text{ on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}\}. \quad (1.61)$$

For

$$0 < s \leq \frac{1}{2} \quad (1.62)$$

the Dirichlet conditions are taken into account with the help of an auxiliary function or at the level of the discrete problem.

1.3.3. Spectral Realization

In the spectral formulation, the fractional operator is prescribed as a fractional power of the local operator in which exchange through the pore boundary has already been taken into account. Let us set

$$\Gamma_D := \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \quad (1.63)$$

$$V_N^{\text{sp}} = \{v \in H^1(D_N) : v = 0 \text{ on } \Gamma_D\}. \quad (1.64)$$

To the local operator

$$A_N u = -\nabla \cdot (a_N \nabla u) + c_N u \quad (1.65)$$

there corresponds the bilinear form

$$\mathbf{a}_N(u, v) = \int_{D_N} a_N \nabla u \cdot \nabla v \, dx + \int_{D_N} c_N u v \, dx + \int_{\Gamma_p(N)} \kappa_N^{\text{sp}} u v \, d\sigma. \quad (1.66)$$

Let

$$A_N \varphi_{m,N} = \lambda_{m,N} \varphi_{m,N}, \quad m = 1, 2, \dots, \quad (1.67)$$

$$(\varphi_{m,N}, \varphi_{k,N})_{L^2(D_N)} = \delta_{mk}. \quad (1.68)$$

Then

$$A_N^s u = \sum_{m=1}^{\infty} \lambda_{m,N}^s (u, \varphi_{m,N})_{L^2(D_N)} \varphi_{m,N}, \quad (1.69)$$

and the corresponding energy form has the form

$$a_{s,N}^{\text{sp}}(u, v) = (A_N^{s/2} u, A_N^{s/2} v)_{L^2(D_N)}. \quad (1.70)$$

To take into account inhomogeneous exchange on the pore boundary, we introduce an auxiliary function $b_N \in V_N^{\text{sp}}$, which satisfies

$$\mathbf{a}_N(b_N, v) = \int_{\Gamma_p(N)} \kappa_N^{\text{sp}} u_{b,N}^{\text{sp}} v \, d\sigma \quad \forall v \in V_N^{\text{sp}}. \quad (1.71)$$

For inhomogeneous Dirichlet conditions on Γ_D , we also fix a function $g_N \in H^1(D_N)$:

$$g_N = U_{\text{in}} \text{ on } \Gamma_{\text{in}}, \quad g_N = U_{\text{out}} \text{ on } \Gamma_{\text{out}}. \quad (1.72)$$

Then we represent the sought solution in the form

$$u_{N,s}^{\text{sp}} = w_{N,s}^{\text{sp}} + g_N + b_N, \quad (1.73)$$

where $w_{N,s}^{\text{sp}}$ is determined from the condition

$$a_{s,N}^{\text{sp}}(w_{N,s}^{\text{sp}}, v) = \int_{D_N} f_N v \, dx - a_{s,N}^{\text{sp}}(g_N + b_N, v) \quad \forall v \in V_{N,s,0}^{\text{sp}}. \quad (1.74)$$

Here $V_{N,s,0}^{\text{sp}} := \mathcal{D}(A_N^{s/2})$, and the zero Dirichlet conditions on Γ_D have already been built into the definition of the operator A_N through the space V_N^{sp} .

1.4. Stationary and Nonstationary Formulations

For all considered diffusion models, it is convenient to use a common notation for the stationary and nonstationary problems.

Let $\text{op} \in \{\text{loc}, \text{int}, \text{reg}, \text{sp}\}$, and let $\mathcal{A}_{N,s}^{(\text{op})}$ denote the corresponding operator for the domain D_N or for the extended interaction domain, if this is provided by the formulation. Here N corresponds to the geometry of the domain, and $s \in (0, 1)$ is the parameter of fractionality.

The stationary problem has the form

$$\mathcal{A}_{N,s}^{(\text{op})} u_{N,s}^{(\text{op})} = F_{N,s}^{(\text{op})}, \quad (1.75)$$

where $u_{N,s}^{(\text{op})}$ is the sought function, and $F_{N,s}^{(\text{op})}$ includes the volumetric source, boundary conditions, and terms describing exchange.

Let $V_{N,s}^{(\text{op})}$ be the space of sought functions, and let $V_{N,s,0}^{(\text{op})}$ be the space of test functions. Then the weak stationary formulation is the following: find $u_{N,s}^{(\text{op})} \in V_{N,s}^{(\text{op})}$ such that

$$a_{N,s}^{(\text{op})}(u_{N,s}^{(\text{op})}, v) + r_{N,s}^{(\text{op})}(u_{N,s}^{(\text{op})}, v) = \ell_{N,s}^{(\text{op})}(v) \quad \forall v \in V_{N,s,0}^{(\text{op})}. \quad (1.76)$$

Here $a_{N,s}^{(\text{op})}(\cdot, \cdot)$ corresponds to the main energy part of the problem, $r_{N,s}^{(\text{op})}(\cdot, \cdot)$ describes exchange through the internal boundaries or the interaction domain, and $\ell_{N,s}^{(\text{op})}$ is the linear functional of the right-hand side. For the spectral formulation, $r_{N,s}^{(\text{sp})} \equiv 0$, since the corresponding exchange has already been taken into account in the form $a_{N,s}^{\text{sp}}$.

The nonstationary problem on the interval $0 < t \leq T$, $T > 0$, is written as

$$\partial_t u_{N,s}^{(\text{op})}(t) + \mathcal{A}_{N,s}^{(\text{op})} u_{N,s}^{(\text{op})}(t) = F_{N,s}^{(\text{op})}(t), \quad t \in (0, T], \quad (1.77)$$

with the initial condition

$$u_{N,s}^{(\text{op})}(0) = u_{0,N}^{(\text{op})} \in V_{N,s}^{(\text{op})}. \quad (1.78)$$

The weak nonstationary formulation has the form: find $u_{N,s}^{(\text{op})} : (0, T) \rightarrow V_{N,s}^{(\text{op})}$ such that

$$(\partial_t u_{N,s}^{(\text{op})}(t), v)_{L^2} + a_{N,s}^{(\text{op})}(u_{N,s}^{(\text{op})}(t), v) + r_{N,s}^{(\text{op})}(u_{N,s}^{(\text{op})}(t), v) = \ell_{N,s}^{(\text{op})}(t; v) \quad (1.79)$$

for all $v \in V_{N,s,0}^{(\text{op})}$ and almost all $t \in (0, T]$.

The stationary problem is a particular case of (1.77), when $\partial_t u_{N,s}^{(\text{op})}(t) \equiv 0$. Thus, in the nonstationary formulation the spatial operator is preserved, and additionally the time term $\partial_t u_{N,s}^{(\text{op})}$ appears.

1.5. Specification of Exchange Through Internal Boundaries

When passing from the domain D_N to another one, not only the geometry changes, but also the measure of the set on which exchange takes place. Therefore, fixing only the pointwise exchange coefficient does not ensure a correct comparison of different domains: with a change of geometry, the total contribution of this term also changes automatically. For this reason, we prescribe the exchange through an integral characteristic.

Let E_N be the set on which exchange takes place. In the considered formulations,

$$E_N = \begin{cases} \Gamma_p(N), & \text{for the local and spectral models,} \\ E_N^{\text{int}} \subset \Lambda_N \setminus D_N, & \text{for the integral model,} \\ \Gamma_p(N), & \text{for the regional model for } s > \frac{1}{2}, \\ (D_N)_\delta, & \text{for the regional model for } s \leq \frac{1}{2}. \end{cases} \quad (1.80)$$

Here

$$(D_N)_\delta = \{x \in D_N : \text{dist}(x, \partial D_N) < \delta\}, \quad \delta > 0. \quad (1.81)$$

On the set E_N , the exchange coefficient is prescribed

$$\kappa_N : E_N \rightarrow [0, +\infty). \quad (1.82)$$

We define the total intensity of exchange by the formula

$$\mathcal{K}_N := \int_{E_N} \kappa_N d\mu, \quad (1.83)$$

where

$$d\mu = \begin{cases} d\sigma, & \text{if exchange is prescribed on the boundary,} \\ dx, & \text{if exchange is prescribed in the volume.} \end{cases} \quad (1.84)$$

Thus, the quantity $\mathcal{K}_N$ characterizes the total strength of exchange on the set E_N .

For the internal boundaries

$$\Gamma_{j,N} = \partial\omega_{j,N}, \quad j = 1, \dots, J_N, \quad (1.85)$$

we introduce component quantities

$$\mathcal{K}_{j,N} \geq 0, \quad (1.86)$$

such that

$$\mathcal{K}_{p,N} = \sum_{j=1}^{J_N} \mathcal{K}_{j,N}. \quad (1.87)$$

A natural distribution has the form

$$\mathcal{K}_{j,N} = \mathcal{K}_{p,N} \frac{\mathcal{H}^{d-1}(\Gamma_{j,N})}{\Sigma_N}, \quad (1.88)$$

where Σ_N is defined in (1.12).

If on $\Gamma_{j,N}$ the exchange coefficient is constant, then

$$\mathcal{K}_{j,N} = \int_{\Gamma_{j,N}} \kappa_{j,N} d\sigma = \kappa_{j,N} \mathcal{H}^{d-1}(\Gamma_{j,N}), \quad (1.89)$$

and therefore

$$\kappa_{j,N} = \frac{\mathcal{K}_{j,N}}{\mathcal{H}^{d-1}(\Gamma_{j,N})}. \quad (1.90)$$

If, however, exchange is prescribed in a volumetric domain

$$E_{j,N} \subset \mathbb{R}^d, \quad (1.91)$$

then, for a constant coefficient,

$$\mathcal{K}_{j,N} = \int_{E_{j,N}} \kappa_{j,N}^{\text{vol}} dx = \kappa_{j,N}^{\text{vol}} |E_{j,N}|, \quad (1.92)$$

from which

$$\kappa_{j,N}^{\text{vol}} = \frac{\mathcal{K}_{j,N}}{|E_{j,N}|}. \quad (1.93)$$

The reason for choosing the quantity (1.83) is seen from the exchange term

$$r_N(u, u) = \int_{E_N} \kappa_N |u|^2 d\mu. \quad (1.94)$$

If on E_N the function u changes little, that is,

$$u|_{E_N} \approx c, \quad c \in \mathbb{R}, \quad (1.95)$$

then

$$r_N(u, u) \approx |c|^2 \int_{E_N} \kappa_N d\mu = |c|^2 \mathcal{K}_N. \quad (1.96)$$

Thus, for slowly varying regimes, the strength of exchange is determined precisely by the integral quantity $\mathcal{K}_N$.

If, on the other hand, one fixes the pointwise coefficient

$$\kappa_N \equiv \kappa_* > 0, \quad (1.97)$$

then

$$r_N(u, u) \approx \kappa_* \mu(E_N) |c|^2. \quad (1.98)$$

Then, with the growth of $\mu(E_N)$, the total exchange is artificially intensified, although the physical meaning of the problem does not change.

Therefore, in what follows exchange is compared by fixed quantities

$$\mathcal{K}_N \quad \text{and} \quad \mathcal{K}_{j,N}, \quad (1.99)$$

and the exchange coefficients themselves are computed from formulas (1.90) and (1.93).

1.6. Integral Characteristics of the Solution

For comparing solutions u , we introduce integral characteristics that are computed directly for all considered models.

1.6.1. Flux Through the Outer Boundary

For the local model, we define the inlet and outlet fluxes by the formulas

$$\Phi_{\text{in},N}^{\text{loc}}(u_N) := - \int_{\Gamma_{\text{in}}} a_N(x) \nabla u_N(x) \cdot \mathbf{n}(x) d\sigma(x), \quad (1.100)$$

$$\Phi_{\text{out},N}^{\text{loc}}(u_N) := \int_{\Gamma_{\text{out}}} a_N(x) \nabla u_N(x) \cdot \mathbf{n}(x) d\sigma(x). \quad (1.101)$$

The sign in (1.100) is chosen so that

$$\Phi_{\text{in},N}^{\text{loc}}(u_N) > 0 \quad (1.102)$$

corresponds to the inflow of flux into the domain.

For the nonlocal models, we prescribe the flux through exchange with a reservoir. If

$$E_{r,N} \quad (1.103)$$

is the corresponding exchange set, then

$$\Phi_{r,N}(u) := \int_{E_{r,N}} \kappa_{r,N}(u_{b,r,N} - u_E) d\mu, \quad (1.104)$$

where

$$d\mu = \begin{cases} d\sigma, & \text{for boundary exchange,} \\ dx, & \text{for volumetric exchange.} \end{cases} \quad (1.105)$$

Here u_E is either the trace of the function u on the boundary, or its restriction to the volumetric set $E_{r,N}$. In this case,

$$\Phi_{r,N}(u) > 0 \quad (1.106)$$

means inflow from the reservoir into the domain.

1.6.2. Total Exchange Through the Pore Boundary

For the internal boundaries

$$\Gamma_{j,N} = \partial\omega_{j,N}, \quad j = 1, \dots, J_N, \quad (1.107)$$

we introduce the total exchange

$$\Phi_{\text{p},N}(u) := \sum_{j=1}^{J_N} \Phi_{j,N}(u). \quad (1.108)$$

It is also convenient to use the quantity of losses through the pores

$$Q_{p,N}(u) := -\Phi_{p,N}(u). \quad (1.109)$$

Then

$$Q_{p,N}(u) \geq 0 \quad (1.110)$$

corresponds to net losses through the internal boundaries.

In the local model, if

$$u_{b,N} = 0 \quad \text{on } \Gamma_p(N), \quad (1.111)$$

then from the Robin condition we have

$$Q_{p,N}^{\text{loc}}(u_N) = \int_{\Gamma_p(N)} \kappa_N u_N d\sigma. \quad (1.112)$$

1.6.3. Normalized Characteristics

Since the total measure of the internal boundaries depends on N , we introduce the normalized quantity

$$\widehat{Q}_{p,N}(u) := \frac{Q_{p,N}(u)}{\Sigma_N}, \quad (1.113)$$

where Σ_N is defined in (1.12). In the two-dimensional case, $\widehat{Q}_{p,N}$ is the average intensity of losses per unit length of the pore boundary.

1.6.4. Effective Conductance

We define the effective conductance as

$$G_{N,s} := \frac{\Phi_{\text{in},N}}{\Delta U}, \quad (1.114)$$

where ΔU is given in (1.22). For fractional operators, we also use the dimensionless characteristic

$$\widehat{G}_{N,s} := L^{2s-2} G_{N,s}, \quad L := \text{diam}(\Omega). \quad (1.115)$$

1.6.5. Relative Losses

We define the fraction of the inlet flux that is lost through the internal boundaries as

$$\mathcal{L}_{N,s} := \frac{Q_{p,N}}{\Phi_{\text{in},N}}, \quad \Phi_{\text{in},N} > 0. \quad (1.116)$$

1.6.6. Comparison with the Local Model

For comparing the nonlocal model with the local one, we introduce the relative deviation

$$\Delta_{N,s} := 100 \frac{\widehat{Q}_{p,N} - \widehat{Q}_{p,N}^{\text{loc}}}{\widehat{Q}_{p,N}^{\text{loc}}}, \quad \widehat{Q}_{p,N}^{\text{loc}} \neq 0. \quad (1.117)$$

The quantity $\Delta_{N,s}$ shows how much the nonlocal model deviates from the local one on the same domain D_N .

1.7. Main Research Problem

Let there be given a set of perforated domains

$$\{D_N\}_{N=0}^{N_{\max}}, \quad (1.118)$$

a fixed potential difference

$$\Delta U = U_{\text{in}} - U_{\text{out}}, \quad U_{\text{in}} > U_{\text{out}}, \quad (1.119)$$

and the law of exchange through the internal boundaries, given by the quantities

$$\mathcal{K}_N^{(\text{op})}, \quad \mathcal{K}_{j,N}. \quad (1.120)$$

For all

$$N = 0, 1, \dots, N_{\max}, \quad s \in (0, 1), \quad \text{op} \in \{\text{loc}, \text{int}, \text{reg}, \text{sp}\} \quad (1.121)$$

it is necessary to find the solution

$$u_{N,s}^{(\text{op})} \quad (1.122)$$

of the corresponding diffusion problem.

After that, the dependencies on the geometric parameter

$$N \longmapsto \widehat{Q}_{\text{p},N}^{(\text{op})}, \quad N \longmapsto G_{N,s}^{(\text{op})}, \quad N \longmapsto \mathcal{L}_{N,s}^{(\text{op})}, \quad (1.123)$$

and on the order of the fractional operator

$$s \longmapsto \widehat{Q}_{\text{p},N}^{(\text{op})}, \quad s \longmapsto G_{N,s}^{(\text{op})}, \quad s \longmapsto \Delta_{N,s}^{(\text{op})}. \quad (1.124)$$

Thus, the main problem consists in the following: under fixed external conditions and consistent internal exchange,

$$\text{construct } u_{N,s}^{(\text{op})} \quad \text{and determine the dependence } (\widehat{Q}_{\text{p},N}^{(\text{op})}, G_{N,s}^{(\text{op})}, \mathcal{L}_{N,s}^{(\text{op})}, \Delta_{N,s}^{(\text{op})}) \quad (1.125)$$

on the geometry of the domain D_N , the order s , and the type of operator op.

Chapter 2

Numerical Solution of Diffusion Problems with Spatially Nonlocal Operators by the Finite Element Method

In this chapter, the finite element approximation of the problems formulated in Chapter 1 is constructed within the framework of the standard variational finite element approach [3, 14].

2.1. Triangulation of the Domain and Finite Element Spaces

For each domain D_N , we consider a conforming triangulation

$$\mathcal{T}_{N,h}(D_N) = \{T\}, \quad (2.1)$$

and we define the mesh parameter as

$$h := \max_{T \in \mathcal{T}_{N,h}(D_N)} \text{diam}(T). \quad (2.2)$$

In the local, regional, and spectral formulations, the unknown function is defined directly in the domain D_N . Therefore, on $\mathcal{T}_{N,h}(D_N)$ we introduce the standard space of continuous piecewise linear functions

$$V_{N,h} := \{v_h \in C(\overline{D_N}) : v_h|_T \in \mathbb{P}_1(T) \quad \forall T \in \mathcal{T}_{N,h}(D_N)\}. \quad (2.3)$$

Here $\mathbb{P}_1(T)$ denotes the space of first-degree polynomials on the element T .

Since on the parts of the boundary Γ_{in} and Γ_{out} Dirichlet conditions are prescribed, we also introduce the subspace

$$V_{N,h,0} := \{v_h \in V_{N,h} : v_h = 0 \text{ on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}\}. \quad (2.4)$$

If

$$g_{N,h} \in V_{N,h} \quad (2.5)$$

is a fixed lifting of the inhomogeneous boundary data, that is,

$$g_{N,h} = U_{\text{in}} \text{ on } \Gamma_{\text{in}}, \quad g_{N,h} = U_{\text{out}} \text{ on } \Gamma_{\text{out}}, \quad (2.6)$$

then any discrete solution can be written in the form

$$u_{N,h} = w_{N,h} + g_{N,h}, \quad w_{N,h} \in V_{N,h,0}. \quad (2.7)$$

For the integral realization, where the unknown function is defined on the extended domain Λ_N , we analogously introduce a conforming triangulation

$$\mathcal{T}_{N,h}(\Lambda_N) \quad (2.8)$$

and the space

$$W_{N,h} := \{w_h \in C(\overline{\Lambda_N}) : w_h|_T \in \mathbb{P}_1(T) \quad \forall T \in \mathcal{T}_{N,h}(\Lambda_N)\}. \quad (2.9)$$

If at the inlet and outlet the Dirichlet conditions are prescribed directly, then the corresponding subspace of test functions has the form

$$W_{N,h,0} := \{w_h \in W_{N,h} : w_h = 0 \text{ on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}\}. \quad (2.10)$$

For the subsequent matrix representations, let us fix the basis functions

$$\{\varphi_i\}_{i=1}^{M_N} \quad (2.11)$$

in the space $V_{N,h,0}$ and, if needed,

$$\{\psi_i\}_{i=1}^{\widetilde{M}_N} \quad (2.12)$$

in the space $W_{N,h,0}$. Then any function from the corresponding finite element space can be represented in the form of a linear combination of these basis functions.

Thus, the subsequent discrete formulations are considered in finite-dimensional subspaces of the spaces $H^1(D_N)$ and $H^1(\Lambda_N)$.

2.2. Discretization of the Local Stationary Problem

The finite element approximation of the local stationary problem has the form: find

$$u_{N,h} \in V_{N,h}, \quad (2.13)$$

such that

$$\begin{aligned} \int_{D_N} a_N \nabla u_{N,h} \cdot \nabla v_h dx + \int_{D_N} c_N u_{N,h} v_h dx + \int_{\Gamma_p(N)} \kappa_N u_{N,h} v_h d\sigma \\ = \int_{D_N} f_N v_h dx + \int_{\Gamma_p(N)} \kappa_N u_{b,N} v_h d\sigma \quad \forall v_h \in V_{N,h,0}. \end{aligned} \quad (2.14)$$

Taking into account the lifting (2.6) and the decomposition (2.7), we set

$$u_{N,h} = w_{N,h} + g_{N,h}, \quad w_{N,h} \in V_{N,h,0}. \quad (2.15)$$

Then we have: find

$$w_{N,h} \in V_{N,h,0}, \quad (2.16)$$

such that

$$A_{N,h}(w_{N,h}, v_h) = L_{N,h}(v_h) \quad \forall v_h \in V_{N,h,0}, \quad (2.17)$$

where

$$A_{N,h}(u_h, v_h) := \int_{D_N} a_N \nabla u_h \cdot \nabla v_h dx + \int_{D_N} c_N u_h v_h dx + \int_{\Gamma_p(N)} \kappa_N u_h v_h d\sigma \quad (2.18)$$

and

$$L_{N,h}(v_h) := \int_{D_N} f_N v_h dx + \int_{\Gamma_p(N)} \kappa_N u_{b,N} v_h d\sigma - A_{N,h}(g_{N,h}, v_h). \quad (2.19)$$

Let

$$w_{N,h} = \sum_{j=1}^{M_N} U_j \varphi_j. \quad (2.20)$$

Substituting (2.20) into (2.17) and choosing successively

$$v_h = \varphi_i, \quad i = 1, \dots, M_N, \quad (2.21)$$

we obtain the matrix system

$$\mathbf{A}^{(N)} \mathbf{U} = \mathbf{F}^{(N)}, \quad (2.22)$$

where

$$\mathbf{A}^{(N)} = \mathbf{K}^{(N)} + \mathbf{C}^{(N)} + \mathbf{R}^{(N)}, \quad (2.23)$$

where

$$K_{ij}^{(N)} = \int_{D_N} a_N \nabla \varphi_j \cdot \nabla \varphi_i \, dx, \quad (2.24)$$

$$C_{ij}^{(N)} = \int_{D_N} c_N \varphi_j \varphi_i \, dx, \quad (2.25)$$

$$R_{ij}^{(N)} = \int_{\Gamma_p(N)} \kappa_N \varphi_j \varphi_i \, d\sigma. \quad (2.26)$$

Theorem 2.2.1 *Let*

$$a_N(x) \geq a_* > 0, \quad c_N(x) \geq 0, \quad \kappa_N(x) \geq 0. \quad (2.27)$$

Then the discrete local problem (2.17) has a unique solution.

Proof. For $v_h \in V_{N,h,0}$, we have

$$\begin{aligned} A_{N,h}(v_h, v_h) &= \int_{D_N} a_N |\nabla v_h|^2 \, dx \\ &\quad + \int_{D_N} c_N |v_h|^2 \, dx \\ &\quad + \int_{\Gamma_p(N)} \kappa_N |v_h|^2 \, d\sigma \\ &\geq a_* \|\nabla v_h\|_{L^2(D_N)}^2. \end{aligned} \quad (2.28)$$

Together with the Poincare inequality, this gives coercivity of the form on $V_{N,h,0}$. Since $V_{N,h,0}$ is finite-dimensional, the discrete problem has a unique solution. $\square$

In the simplest basic case

$$a_N \equiv 1, \quad c_N \equiv 0, \quad f_N \equiv 0, \quad (2.29)$$

the discrete problem simplifies to the form

$$\begin{aligned} \int_{D_N} \nabla u_{N,h} \cdot \nabla v_h \, dx + \int_{\Gamma_p(N)} \kappa_N u_{N,h} v_h \, d\sigma \\ = \int_{\Gamma_p(N)} \kappa_N u_{b,N} v_h \, d\sigma, \quad (2.30) \\ \forall v_h \in V_{N,h,0}. \end{aligned}$$

This variant is used in what follows as a reference one compared with the spatially nonlocal models.

2.3. Discretization of Spatially Nonlocal Models

In this subsection, we construct finite-dimensional approximations of the integral, regional, and spectral realizations. For the local and regional models, we use the spaces $V_{N,h}$ and $V_{N,h,0}$, and for the integral model the spaces $W_{N,h}$ and $W_{N,h,0}$.

Integral Realization

For the integral realization, we use the spaces $W_{N,h}$ and $W_{N,h,0}$ introduced in subsection 2.1.. Such a type of discretization is consistent with known finite element approaches to problems with the fractional Laplacian [1].

We seek

$$u_{N,s,h}^{\text{int}} \in W_{N,h} \quad (2.31)$$

from the condition

$$a_{s,N,h}^{\text{int}}(u_{N,s,h}^{\text{int}}, v_h) + r_{s,N,h}^{\text{int}}(u_{N,s,h}^{\text{int}}, v_h) = \ell_{s,N,h}^{\text{int}}(v_h) \quad \forall v_h \in W_{N,h,0}, \quad (2.32)$$

where

$$\begin{aligned} a_{s,N,h}^{\text{int}}(u_h, v_h) &:= \frac{1}{2} \iint_{\Lambda_N \times \Lambda_N} (u_h(x) - u_h(y))(v_h(x) - v_h(y)) \\ &\quad \times K_s(x, y) \, dx \, dy, \end{aligned} \quad (2.33)$$

$$r_{s,N,h}^{\text{int}}(u_h, v_h) := \int_{E_N^{\text{int}}} \kappa_N^{\text{int}} u_h v_h \, dx, \quad (2.34)$$

$$\ell_{s,N,h}^{\text{int}}(v_h) := \int_{D_N} f_N v_h \, dx + \int_{E_N^{\text{int}}} \kappa_N^{\text{int}} u_{b,N}^{\text{int}} v_h \, dx. \quad (2.35)$$

If

$$u_{N,s,h}^{\text{int}} = \sum_{j=1}^{\widetilde{M}_N} U_j \psi_j, \quad v_h = \psi_i, \quad (2.36)$$

then we obtain the system

$$\sum_{j=1}^{\widetilde{M}_N} (A_{ij}^{\text{int}} + R_{ij}^{\text{int}}) U_j = F_i^{\text{int}}, \quad i = 1, \dots, \widetilde{M}_N, \quad (2.37)$$

where

$$A_{ij}^{\text{int}} = \frac{1}{2} \iint_{\Lambda_N \times \Lambda_N} (\psi_j(x) - \psi_j(y))(\psi_i(x) - \psi_i(y)) K_s(x, y) \, dx \, dy, \quad (2.38)$$

$$R_{ij}^{\text{int}} = \int_{E_N^{\text{int}}} \kappa_N^{\text{int}} \psi_j \psi_i \, dx, \quad (2.39)$$

$$F_i^{\text{int}} = \int_{D_N} f_N \psi_i \, dx + \int_{E_N^{\text{int}}} \kappa_N^{\text{int}} u_{b,N}^{\text{int}} \psi_i \, dx. \quad (2.40)$$

Regional Realization

For the regional realization, we use the spaces $V_{N,h}$ and $V_{N,h,0}$.

We seek

$$u_{N,s,h}^{\text{reg}} \in V_{N,h} \quad (2.41)$$

from the condition

$$a_{s,N,h}^{\text{reg}}(u_{N,s,h}^{\text{reg}}, v_h) + r_{s,N,h}^{\text{reg},*}(u_{N,s,h}^{\text{reg}}, v_h) = \ell_{s,N,h}^{\text{reg},*}(v_h) \quad \forall v_h \in V_{N,h,0}, \quad (2.42)$$

where

$$a_{s,N,h}^{\text{reg}}(u_h, v_h) := \frac{1}{2} \iint_{D_N \times D_N} (u_h(x) - u_h(y))(v_h(x) - v_h(y)) K_s(x, y) \, dx \, dy. \quad (2.43)$$

If

$$s > \frac{1}{2}, \quad (2.44)$$

then

$$r_{s,N,h}^{\text{reg}}(u_h, v_h) = \int_{\Gamma_p(N)} \kappa_N^{\text{reg}} u_h v_h d\sigma, \quad (2.45)$$

$$\ell_{s,N,h}^{\text{reg}}(v_h) = \int_{D_N} f_N v_h dx + \int_{\Gamma_p(N)} \kappa_N^{\text{reg}} u_{b,N}^{\text{reg}} v_h d\sigma. \quad (2.46)$$

If

$$s \leq \frac{1}{2}, \quad (2.47)$$

then

$$r_{s,N,h}^{\text{lay}}(u_h, v_h) = \int_{(D_N)_\delta} \kappa_{N,\delta}^{\text{lay}} u_h v_h dx, \quad (2.48)$$

$$\ell_{s,N,h}^{\text{lay}}(v_h) = \int_{D_N} f_N v_h dx + \int_{(D_N)_\delta} \kappa_{N,\delta}^{\text{lay}} u_{b,N,\delta}^{\text{lay}} v_h dx. \quad (2.49)$$

In the basis

$$u_{N,s,h}^{\text{reg}} = \sum_{j=1}^{M_N} U_j \varphi_j \quad (2.50)$$

we obtain the system

$$\sum_{j=1}^{M_N} \left(A_{ij}^{\text{reg}} + R_{ij}^{\text{reg},*} \right) U_j = F_i^{\text{reg},*}, \quad i = 1, \dots, M_N, \quad (2.51)$$

where

$$A_{ij}^{\text{reg}} = \frac{1}{2} \iint_{D_N \times D_N} (\varphi_j(x) - \varphi_j(y)) (\varphi_i(x) - \varphi_i(y)) K_s(x, y) dx dy. \quad (2.52)$$

Spectral Realization

For the spectral model, in the space $V_{N,h,0}$ we introduce the matrices

$$K_{ij}^{(N)} = \int_{D_N} a_N \nabla \varphi_j \cdot \nabla \varphi_i dx + \int_{D_N} c_N \varphi_j \varphi_i dx + \int_{\Gamma_p(N)} \kappa_N^{\text{sp}} \varphi_j \varphi_i d\sigma, \quad (2.53)$$

$$M_{ij}^{(N)} = \int_{D_N} \varphi_j \varphi_i dx. \quad (2.54)$$

Further, we consider the generalized spectral problem

$$\mathbf{K}^{(N)} \mathbf{z}_m = \lambda_{m,N,h} \mathbf{M}^{(N)} \mathbf{z}_m, \quad m = 1, \dots, M_N, \quad (2.55)$$

where we normalize the eigenvectors by the condition

$$(\mathbf{z}_m)^T \mathbf{M}^{(N)} \mathbf{z}_k = \delta_{mk}. \quad (2.56)$$

The discrete fractional power of the operator is prescribed as [2]

$$A_{N,h}^s u_h = \sum_{m=1}^{M_N} \lambda_{m,N,h}^s (u_h, \varphi_{m,h})_{L^2(D_N)} \varphi_{m,h}, \quad (2.57)$$

where $\varphi_{m,h}$ is the finite element function corresponding to the vector $\mathbf{z}_m$.

We seek

$$u_{N,s,h}^{\text{sp}} = w_{N,s,h}^{\text{sp}} + g_{N,h} + b_{N,h}, \quad (2.58)$$

where for $w_{N,s,h}^{\text{sp}}$ we have

$$a_{s,N,h}^{\text{sp}}(w_{N,s,h}^{\text{sp}}, v_h) = \int_{D_N} f_N v_h dx - a_{s,N,h}^{\text{sp}}(g_{N,h} + b_{N,h}, v_h) \quad (2.59)$$

$$\forall v_h \in V_{N,s,0}^{\text{sp}},$$

where

$$a_{s,N,h}^{\text{sp}}(u_h, v_h) := (A_{N,h}^{s/2} u_h, A_{N,h}^{s/2} v_h)_{L^2(D_N)}. \quad (2.60)$$

Theorem 2.3.1 *The matrices of the discrete local and spectral problems are symmetric. For the integral and regional realizations, symmetry is also preserved under symmetric computation of the double integrals.*

Proof. In all cases, the bilinear forms are defined through symmetric expressions with respect to permutation of the arguments u and v . In particular,

$$(u(x) - u(y))(v(x) - v(y)) = (v(x) - v(y))(u(x) - u(y)), \quad (2.61)$$

and the boundary and volumetric exchange terms have the form of the product uv . Therefore, the corresponding matrices are symmetric. $\square$

2.4. Discretization of the Nonstationary Problem

For the nonstationary problem, after spatial discretization we obtain the semidiscrete system

$$\mathbf{M} \frac{d\mathbf{U}}{dt}(t) + \mathbf{A} \mathbf{U}(t) = \mathbf{F}(t), \quad 0 < t \leq T, \quad (2.62)$$

where

$$\mathbf{M} = (M_{ij}), \quad M_{ij} = \int_{\Omega_h} \phi_j \phi_i dx, \quad (2.63)$$

and the matrix $\mathbf{A}$ corresponds to the chosen spatial formulation:

$$\mathbf{A} = \mathbf{A}^{\text{loc}}, \quad \mathbf{A}^{\text{int}}, \quad \mathbf{A}^{\text{reg}}, \quad \text{or} \quad \mathbf{A}^{\text{sp}}. \quad (2.64)$$

Here Ω_h denotes D_N or Λ_N depending on the model.

Let

$$0 = t_0 < t_1 < \dots < t_M = T, \quad \tau_n = t_{n+1} - t_n. \quad (2.65)$$

For the time approximation, we use the θ -scheme:

$$\mathbf{M} \frac{\mathbf{U}^{n+1} - \mathbf{U}^n}{\tau_n} + \mathbf{A}(\theta \mathbf{U}^{n+1} + (1 - \theta) \mathbf{U}^n) = \theta \mathbf{F}^{n+1} + (1 - \theta) \mathbf{F}^n, \quad (2.66)$$

where

$$\theta \in [0, 1]. \quad (2.67)$$

From here we obtain the matrix system at each time step:

$$(\mathbf{M} + \tau_n \theta \mathbf{A}) \mathbf{U}^{n+1} = (\mathbf{M} - \tau_n (1 - \theta) \mathbf{A}) \mathbf{U}^n + \tau_n (\theta \mathbf{F}^{n+1} + (1 - \theta) \mathbf{F}^n). \quad (2.68)$$

Particular cases of (2.68) are:

$$\theta = 1 \quad \implies \quad (\mathbf{M} + \tau_n \mathbf{A}) \mathbf{U}^{n+1} = \mathbf{M} \mathbf{U}^n + \tau_n \mathbf{F}^{n+1}, \quad (2.69)$$

$$\theta = \frac{1}{2} \quad \implies \quad \left(\mathbf{M} + \frac{\tau_n}{2} \mathbf{A} \right) \mathbf{U}^{n+1} = \left(\mathbf{M} - \frac{\tau_n}{2} \mathbf{A} \right) \mathbf{U}^n + \frac{\tau_n}{2} (\mathbf{F}^{n+1} + \mathbf{F}^n). \quad (2.70)$$

The initial condition is prescribed in the form

$$\mathbf{U}^0 = \mathbf{U}_{0,h}, \quad (2.71)$$

where $\mathbf{U}_{0,h}$ is the vector of coefficients of the finite element approximation of the initial function $u_{0,N}$.

Theorem 2.4.1 *Let $\mathbf{M}$ be a symmetric positive definite matrix, and let $\mathbf{A}$ be a symmetric nonnegative definite matrix. Then for*

$$\theta \geq \frac{1}{2} \quad (2.72)$$

the scheme (2.68) is unconditionally stable.

Proof. Taking in (2.66) as the test vector $\mathbf{U}^{n+1} - \mathbf{U}^n$, we obtain the standard energy estimate

$$\frac{1}{2} \left(\|\mathbf{U}^{n+1}\|_{\mathbf{M}}^2 - \|\mathbf{U}^n\|_{\mathbf{M}}^2 \right) + \left(\theta - \frac{1}{2} \right) \|\mathbf{U}^{n+1} - \mathbf{U}^n\|_{\mathbf{A}}^2 \leq \tau_n (\text{right-hand side contribution}), \quad (2.73)$$

where

$$\|\mathbf{v}\|_{\mathbf{M}}^2 = (\mathbf{M}\mathbf{v}, \mathbf{v}), \quad \|\mathbf{v}\|_{\mathbf{A}}^2 = (\mathbf{A}\mathbf{v}, \mathbf{v}). \quad (2.74)$$

For $\theta \geq \frac{1}{2}$, the second term is nonnegative, from which unconditional stability follows. $\square$

2.5. Basic Properties of the Discrete Schemes

The constructed discrete schemes have the following properties.

For the local stationary problem, the discrete bilinear form is symmetric and coercive, therefore the corresponding system of linear algebraic equations has a unique solution.

For the integral and regional models, under accurate approximation of the double integrals, the discrete bilinear forms preserve the symmetry of the continuous formulations.

For the nonstationary problem, after spatial discretization a matrix system of the form is obtained

$$\mathbf{M} \frac{d\mathbf{U}}{dt}(t) + \mathbf{A} \mathbf{U}(t) = \mathbf{F}(t), \quad (2.75)$$

for which, for $\theta \geq \frac{1}{2}$, the scheme (2.68) is unconditionally stable.

In the spectral realization, truncation of the expansion in eigenfunctions leads to an additional error, which is determined by the number of retained modes.

Thus, the constructed schemes reduce the continuous problems to finite-dimensional systems suitable for stationary and nonstationary numerical computation.

Chapter 3

Software for Modeling and Analysis of Results

In this chapter, the software developed for numerical modeling of diffusion problems with local and spatially nonlocal operators is described, together with the tools for processing, visualization, and analysis of the results. All main components of the system were implemented by the author in C++.

The software is built as a unified computational system that combines geometry construction, mesh generation, assembly of discrete operators, solution of stationary and nonstationary problems, and visualization of results using the graphics processor. Such a structure corresponds to the typical stages of variational finite element modeling [3, 14].

3.1. General Characteristic of the Software

The software is intended for solving local, integral, regional, and spectral diffusion models in perforated domains with complex internal geometry.

The main functions of the software can be written as

$$\mathcal{S} = \mathcal{S}_{\text{geom}} \cup \mathcal{S}_{\text{mesh}} \cup \mathcal{S}_{\text{disc}} \cup \mathcal{S}_{\text{solve}} \cup \mathcal{S}_{\text{post}}, \quad (3.1)$$

where $\mathcal{S}_{\text{geom}}$ is responsible for geometry construction, $\mathcal{S}_{\text{mesh}}$ for finite element mesh generation, $\mathcal{S}_{\text{disc}}$ for discretization of the operators, $\mathcal{S}_{\text{solve}}$ for solving the discrete systems, and $\mathcal{S}_{\text{post}}$ for processing, analysis, and visualization of the results.

Each computational experiment within the software is determined by a set of parameters

$$\mathcal{P} = (N, s, \text{op}, h, \Delta U, \kappa, u_b, T, \tau), \quad (3.2)$$

where N specifies the domain, s is the order of the fractional operator, op is the model type, h is the mesh parameter, ΔU is the macroscopic load, κ and u_b are the exchange parameters, and T and τ are used in nonstationary computations. The system architecture is oriented toward serial execution of computational experiments, while the visualization subsystem uses the graphics processor to display the geometry, the mesh, and the solutions.

3.2. Structure of the Software

Logically, the software consists of several interconnected modules.

The geometry construction module forms the domain

$$D_N = \Omega \setminus \bigcup_{j=1}^{J_N} \overline{\omega_{j,N}}, \quad (3.3)$$

and partitions its boundary into the parts

$$\Gamma_{\text{in}}, \quad \Gamma_{\text{out}}, \quad \Gamma_{\text{ins}}, \quad \Gamma_{\text{p}}(N). \quad (3.4)$$

At this stage, the geometric parameters of the domain, the number of internal pores, their arrangement, and auxiliary characteristics used later during mesh generation and the prescription of boundary conditions are specified.

The geometry construction module supports both explicitly prescribed domains and parametric generation of families of domains with variable internal structure. It makes it possible to build multiply connected domains with

an arbitrary number of internal cavities, prescribe boundaries analytically, parametrically, or algorithmically, and generate self-similar and fractal-type geometries.

The mesh generation module builds the triangulation

$$\mathcal{T}_{N,h}(D_N) \quad (3.5)$$

or, in the case of the integral model,

$$\mathcal{T}_{N,h}(\Lambda_N). \quad (3.6)$$

The result is a discrete geometry consistent with the mathematical formulation of the problem and a set of nodes, elements, boundary edges, and auxiliary data structures required for matrix assembly.

The mesh generator supports the construction of triangulations for geometries with a large number of internal boundaries. It is based on optimized Delaunay triangulation, suitable for automatic mesh generation in parametrically prescribed domains, self-similar configurations, and geometries with narrow channels and small elements. Local mesh refinement is provided near internal boundaries, corners, and regions with more complicated solution behavior. The current implementation is not positioned as fully parallel; however, its structure is suitable for further parallelization at the stages of generation and processing of independent mesh fragments.

The discretization module realizes the construction of matrices and right-hand-side vectors for all considered models. In general, at this stage the system

$$\mathbf{A}\mathbf{u} = \mathbf{f} \quad (3.7)$$

is formed for the stationary problem or

$$\mathbf{M}\frac{d\mathbf{u}}{dt} + \mathbf{A}\mathbf{u} = \mathbf{f}(t) \quad (3.8)$$

for the nonstationary formulation.

The solution module is responsible for numerical computation of the vector of unknowns

$$\mathbf{u} = (u_1, \dots, u_m)^T, \quad (3.9)$$

which contains the values of the finite element approximation at the nodes or the coefficients of expansion in basis functions.

The analytical module computes the integral characteristics introduced in Chapter 1, in particular

$$Q_{\mathbf{p},N}^{(\text{op})}, \quad \widehat{Q}_{\mathbf{p},N}^{(\text{op})}, \quad G_{N,s}^{(\text{op})}, \quad \mathcal{L}_{N,s}^{(\text{op})}, \quad \Delta_{N,s}^{(\text{op})}, \quad (3.10)$$

and also produces plots, heat maps, solution surfaces, and summary tables for further analysis.

For a concise description of the interaction of the modules, one can write

$$\mathbf{u} = \mathcal{M}_{\text{solve}} \left(\mathcal{M}_{\text{disc}} \left(\mathcal{M}_{\text{mesh}} \left(\mathcal{M}_{\text{geom}}(\mathcal{P}) \right) \right) \right), \quad (3.11)$$

where

$$\mathcal{M}_{\text{geom}}, \quad \mathcal{M}_{\text{mesh}}, \quad \mathcal{M}_{\text{disc}}, \quad \mathcal{M}_{\text{solve}} \quad (3.12)$$

denote the corresponding stages of the computational process.

3.3. Main Capabilities of the System

The developed software supports the construction of local, integral, regional, and spectral models within a unified environment, parametric generation of multiply connected domains, mesh generation based on optimized Delaunay triangulation, solution of stationary and nonstationary problems, computation of integral characteristics, and automated storage of results. A separate emphasis is placed on efficient visualization: rendering of the geometry, the mesh, two-dimensional maps, and three-dimensional solution surfaces is performed using the graphics processor, which enables interactive work on different devices with a compatible graphics subsystem.

From a practical point of view, this means that for a given set of parameters (3.2) the user can perform the full calculation cycle: from prescribing the parameters $\mathcal{P}$ and constructing the domain D_N , through generating the mesh $\mathcal{T}_{N,h}$ and forming the discrete problem $(\mathbf{A}, \mathbf{M}, \mathbf{f})$, to computing the solution $\mathbf{u}$, numerical characteristics, and graphical materials.

For each run, the problem parameters, the operator type, geometric characteristics, mesh parameters, and numerical results are stored. Owing to this, any numerical experiment can be reproduced again with the same set of input data.

3.4. Visualization and Analysis of Results

An important component of the software is the visualization subsystem, since it makes it possible not only to obtain the numerical solution but also to analyze quickly its behavior depending on the geometry of the domain, the operator type, and the value of the parameter s .

All main visualizations are performed using the graphics processor. The software supports rendering of the domain geometry, the finite element mesh, two-dimensional solution maps, three-dimensional surfaces, dependence plots, and comparative diagrams. This approach reduces the load on the central processor during interactive work and makes it possible to run the system on different devices with a compatible graphics subsystem.

For each solution, the quantities

$$u_h(x, y), \quad Q_{p,N}^{(\text{op})}, \quad \widehat{Q}_{p,N}^{(\text{op})}, \quad G_{N,s}^{(\text{op})}, \quad \mathcal{L}_{N,s}^{(\text{op})} \quad (3.13)$$

can be plotted as functions of N , s , or the experiment number.

Figure 3.1 shows the main window of the software. It contains the tools for specifying the problem parameters, selecting the model, starting the computation, and viewing the current state of the experiment. The layout is organized so

that the user can move from problem setup to analysis of the results without changing the working environment.

Figure 3.2 shows the finite element mesh visualization mode for the domain D_N . This view makes it possible to verify the consistency of the discrete geometry with the original domain, analyze the shape and size of the elements, and monitor mesh refinement near internal boundaries and geometrically more complicated regions. Figure 3.3 shows an example of displaying the numerical solution as a field of values on the computational domain. Such a visualization makes it possible to trace directly the spatial distribution of the sought quantity, identify local features near pores and the outer boundary, and compare results for different operator types.

Figure 3.4 shows the tools for analyzing the integral characteristics. In this mode, the results are presented in the form of plots, comparative dependencies, and summary numerical data, which makes it possible to analyze systematically the influence of the parameters N , s , and the operator type on the behavior of the solution. The combination of numerical computation, visual control, and analytical processing within a single interface substantially simplifies interpretation of the results and makes the software a convenient tool for research work.

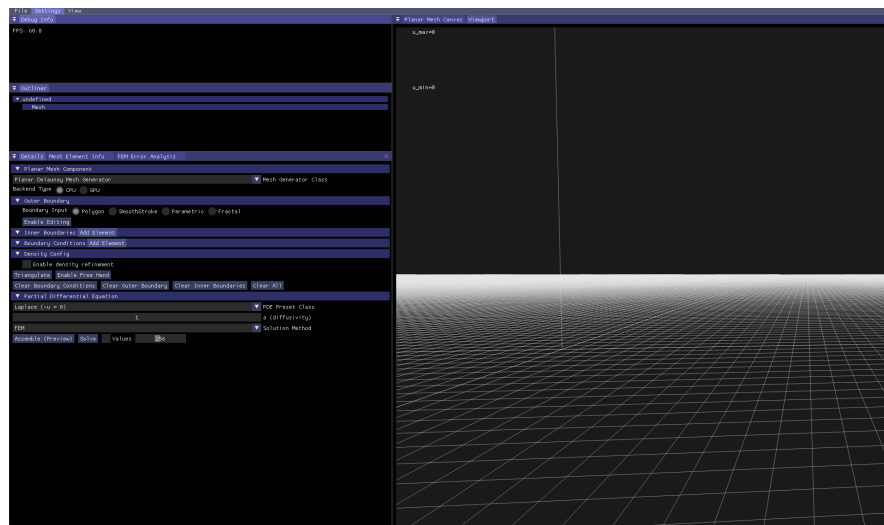


Figure 3.1: Main window of the software.

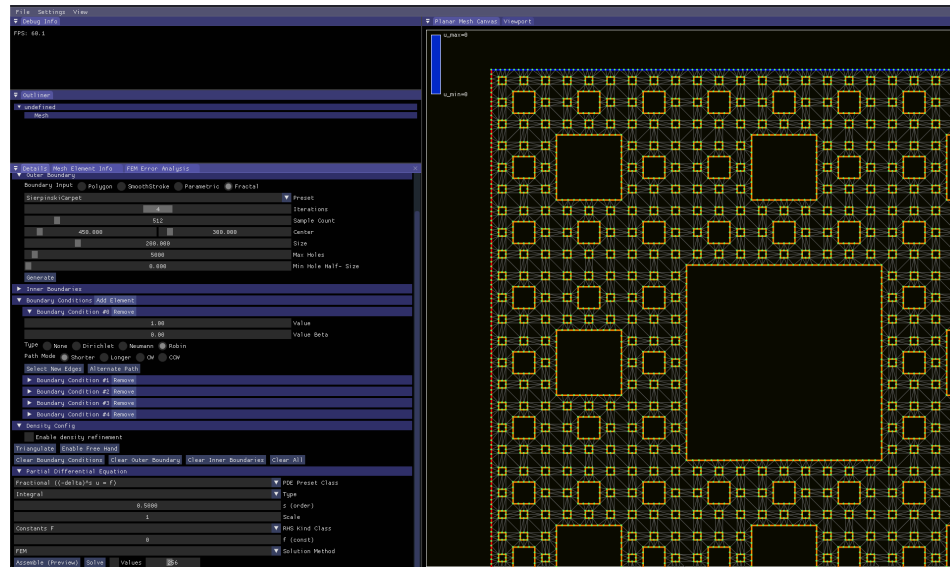


Figure 3.2: Finite element mesh of the domain.

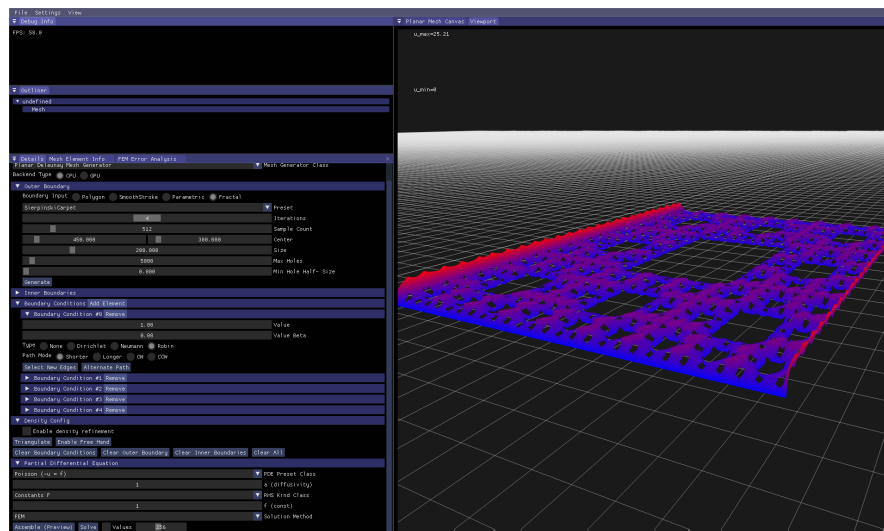


Figure 3.3: Field of the numerical solution.

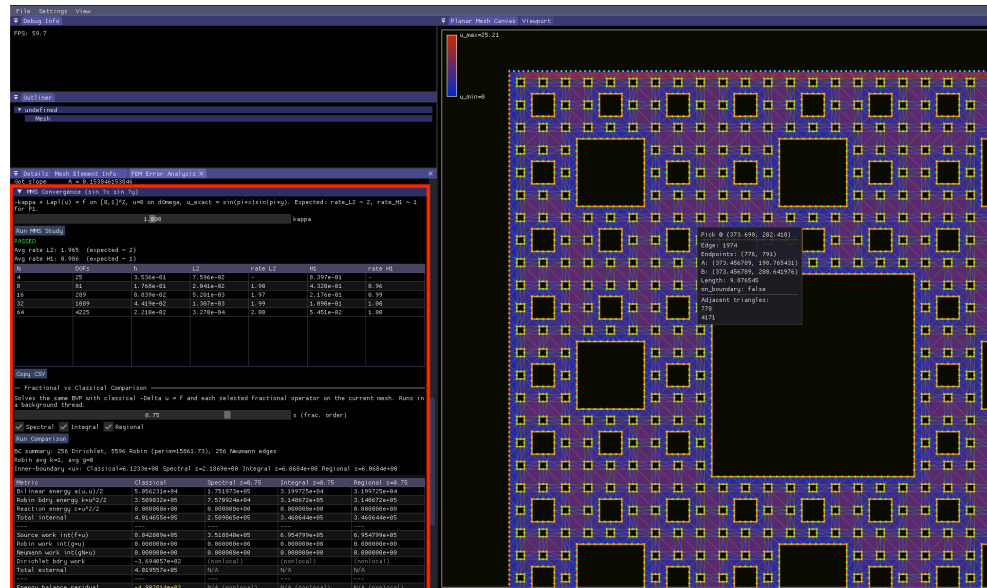


Figure 3.4: Analysis of integral characteristics.

Chapter 4

Numerical Experiments

Now let us present the results of numerical experiments for the local, integral, regional, and spectral diffusion models, as well as the visual analysis of the solution fields for different geometries of the domain, different values of the fractional parameter s , and different operator realizations.

4.1. Scheme of Computational Experiments

Each computational experiment is determined by a set of parameters

$$\mathcal{P} = (D_N, s, \text{op}, h, \Delta U, \kappa, u_b, T, \tau), \quad (4.1)$$

where D_N is the chosen domain, $s \in (0, 1)$ is the fractionality parameter, op is the type of operator, h is the mesh parameter, ΔU is the external potential difference, κ and u_b are the parameters of exchange through the internal boundaries, and T and τ are used in the nonstationary case.

For the analysis, the quantities

$$Q_{\text{p},N}^{(\text{op})}, \quad \widehat{Q}_{\text{p},N}^{(\text{op})}, \quad G_{N,s}^{(\text{op})}, \quad \mathcal{L}_{N,s}^{(\text{op})}, \quad \Delta_{N,s}^{(\text{op})}, \quad (4.2)$$

introduced in Chapter 1, are used.

In all experiments, the comparison is carried out under the same macroscopic load and under equally prescribed exchange through the internal boundaries.

4.2. Results of Numerical Experiments

Let the domains D_N , constructed after the motif of the Sierpinski carpet [7], be used for the values

$$N = 1, 2, 3, 4. \quad (4.3)$$

Exactly such a series of comparative experiments is consistent with the formulation proposed in [11]. For each domain, the same potential difference is considered

$$\Delta U = U_{\text{in}} - U_{\text{out}}, \quad (4.4)$$

and the exchange parameters through the pore boundary are prescribed in a consistent form.

For each domain, a finite element mesh is built, after which computations are performed for the local, integral, regional, and spectral models.

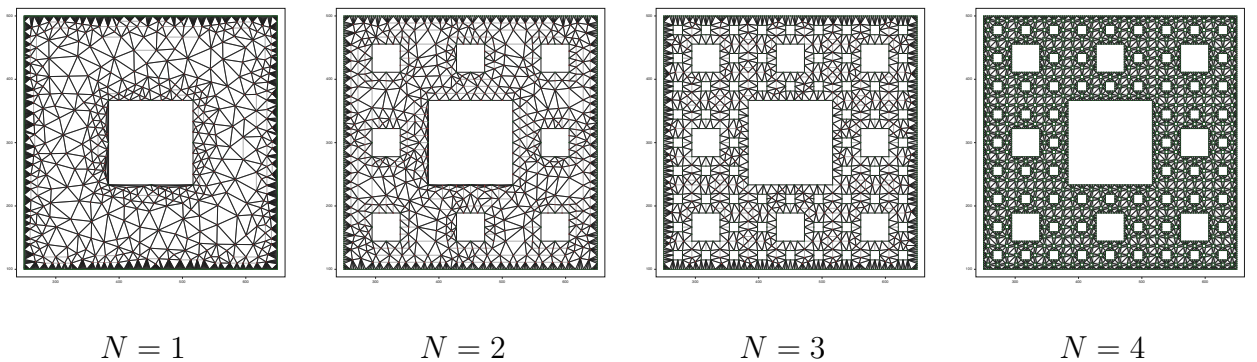


Figure 4.1: An example of a finite element mesh for domains of the Sierpinski carpet type.

From Figure 4.1 it can be seen that as N increases, the geometry simultaneously acquires a larger number of internal pores and a greater total length of the pore boundary. For numerical solution, this means two important

circumstances. First, the configuration of the domain itself becomes more complicated: the number of local narrowings, narrow channels, and regions where the solution field changes more rapidly increases. Second, the contribution of internal exchange through the boundary $\Gamma_p(N)$ grows, and therefore the geometry begins to influence the integral transport characteristics more substantially. For this reason, the domain D_4 is illustrative for comparing operator realizations: it already contains a sufficiently developed internal structure for the differences between the local, integral, regional, and spectral models to be not merely formal, but also noticeable at the level of the computed fluxes and spatial fields.

For the most detailed domain D_4 , for which we have a full report of numerical results, we compare first of all the losses through the pore boundary

$$Q_{p,4}^{(\text{op})}, \quad \widehat{Q}_{p,4}^{(\text{op})}, \quad \Delta_{4,s}^{(\text{op})}. \quad (4.5)$$

Here $\text{op} \in \{\text{loc}, \text{sp}, \text{int}, \text{reg}\}$ denotes respectively the local, spectral, integral, and regional realizations of the operator, and $s \in (0, 1)$ is the order of the fractional operator. The quantity $Q_{p,4}^{(\text{op})}$ is the total flux through the pore boundary of the domain D_4 , and

$$\widehat{Q}_{p,4}^{(\text{op})} := \frac{Q_{p,4}^{(\text{op})}}{\Sigma_4} \quad (4.6)$$

is the normalized flux, where Σ_4 is the total length of the pore boundary. The relative deviation from the local model is prescribed by the formula

$$\Delta_{4,s}^{(\text{op})} := \frac{\widehat{Q}_{p,4}^{(\text{op})} - \widehat{Q}_{p,4}^{(\text{loc})}}{\widehat{Q}_{p,4}^{(\text{loc})}} \cdot 100\%. \quad (4.7)$$

For the local model, we have the reference value

$$\widehat{Q}_{p,4}^{(\text{loc})} = 6.1233. \quad (4.8)$$

Table 4.1: Normalized losses through the pore boundary for the domain D_4 and the relative deviation from the local model.

s	$\widehat{Q}_{p,4}^{(\text{sp})}$	$\widehat{Q}_{p,4}^{(\text{int})}$	$\widehat{Q}_{p,4}^{(\text{reg})}$	$\Delta_{4,s}^{(\text{sp})}, \%$	$\Delta_{4,s}^{(\text{int})}, \%$	$\Delta_{4,s}^{(\text{reg})}, \%$
0.50	1.233	5.899	5.899	-79.9	-3.66	-3.66
0.60	1.550	5.990	5.990	-74.7	-2.19	-2.19
0.75	2.187	6.068	6.068	-64.3	-0.90	-0.90
0.95	3.463	6.122	6.122	-43.4	-0.02	-0.02

The integral and regional realizations for this geometry practically coincide. The greatest deviation from the local model is given by the spectral realization.

Table 4.1 makes it possible to see not only the order of magnitude of the quantities, but also the character of the deviation of each model from the reference local formulation. For the integral and regional realizations, the normalized flux remains very close to the local value over the entire interval of s , and as s increases this difference systematically decreases. This agrees with the expectation that when $s \rightarrow 1$, spatially nonlocal diffusion approaches the local model, and therefore the macroscopic intensity of transport through the pore boundary becomes almost the same as in the classical model.

The spectral realization demonstrates a different picture. For small and moderate values of s , it yields substantially smaller values of $\widehat{Q}_{p,4}^{(\text{sp})}$ than the local model. In the physical interpretation, this means that the spectral operator leads to a smoother distribution of potential in the domain and, accordingly, to a less intensive redistribution of flux toward the internal boundaries. As s increases, this effect weakens, but even at $s = 0.95$ the spectral realization still differs noticeably from the local one. Therefore, in this series of experiments, the spectral model turns out to be the most sensitive to the choice of the fractional order.

For the fixed domain D_4 , a series of computations is performed for

$$s \in \{0.5, 0.6, 0.75, 0.95\}, \quad (4.9)$$

and also for the local model as the reference case.

We also consider the total internal energy

$$\mathcal{E}_{4,s}^{(\text{op})} := \mathcal{E}_{a,4,s}^{(\text{op})} + \mathcal{E}_{R,4,s}^{(\text{op})}, \quad (4.10)$$

where $\mathcal{E}_{a,4,s}^{(\text{op})}$ is the main energetic component, and $\mathcal{E}_{R,4,s}^{(\text{op})}$ is the exchange component on the pore boundary. In the full numerical report, a convergence table for the local model problem and auxiliary characteristics $u_{\min}$, $u_{\max}$, the work of the source, and absolute fluxes are also given. In the text of the diploma thesis, only the quantities that are used directly for comparison of the operator realizations are retained.

At the same time, the dependencies

$$s \mapsto \widehat{Q}_{p,4}^{(\text{op})}, \quad s \mapsto \mathcal{E}_{4,s}^{(\text{op})}, \quad s \mapsto \Delta_{4,s}^{(\text{op})}. \quad (4.11)$$

are analyzed.

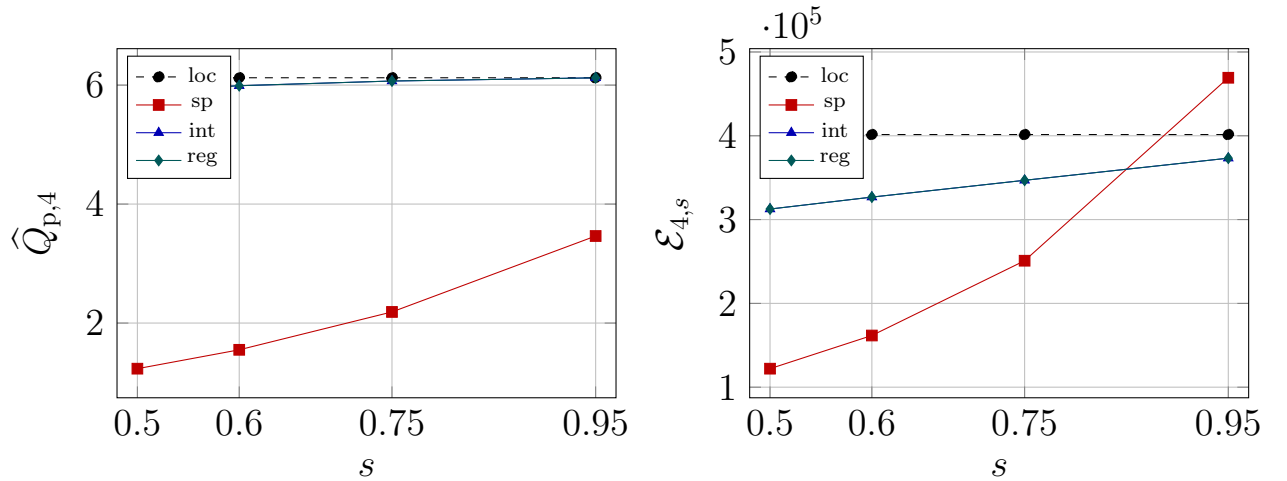


Figure 4.2: Dependencies of $\widehat{Q}_{p,4}^{(\text{op})}$ and $\mathcal{E}_{4,s}^{(\text{op})}$ on the parameter s for the domain D_4 . The dashed line corresponds to the local model.

For the spectral realization, $\Delta_{4,0.5}^{(\text{sp})} \approx -79.9\%$ and $\Delta_{4,0.95}^{(\text{sp})} \approx -43.4\%$. For the integral and regional realizations, the deviation does not exceed a few percent and at $s = 0.95$ practically vanishes.

The graphs in Figure 4.2 clarify this tendency. The left diagram shows that for the integral and regional models the flux curve runs almost along the local reference line, and the deviations are rather of a corrective character than of a qualitatively different nature. For the spectral realization, the dependence on s is much more pronounced: as the fractional order increases, the flux grows monotonically, but the approach to the local horizontal line proceeds slowly. Consequently, in the spectral model the parameter s acts as a substantial regulator of the transport intensity, whereas in the integral and regional formulations its influence in this geometry is milder.

The right diagram for $\mathcal{E}_{4,s}^{(\text{op})}$ confirms the same conclusion at the energetic level. The local model sets the highest level of total energy, the integral and regional realizations occupy an intermediate position, and the spectral model for small s gives the smallest values. This means that the differences between the realizations manifest themselves not only in the integral flux through the pore boundary, but also in the overall spatial structure of the field, that is, in how strongly the solution changes throughout the domain and how intensively the internal gradient transport is formed.

The two-dimensional maps in Figures 4.3–4.6 are now presented separately for each value of s , so the differences between the realizations are visible much more clearly. In the local model, in all four cases the characteristic gradient structure is preserved: the potential changes from inlet to outlet in a nonuniform way, and the pore geometry produces local deformations of the field near the internal boundaries. It is clearly visible that even under the same external loading, the complex internal structure of the domain forms a network of channels in which the distribution of values is not uniform.

In Figure 4.3, which corresponds to $s = 0.5$, the differences between the models are the most pronounced. The integral and regional realizations give a noticeably higher level of the field in a larger part of the domain than the local model, although their spatial patterns are almost indistinguishable from each

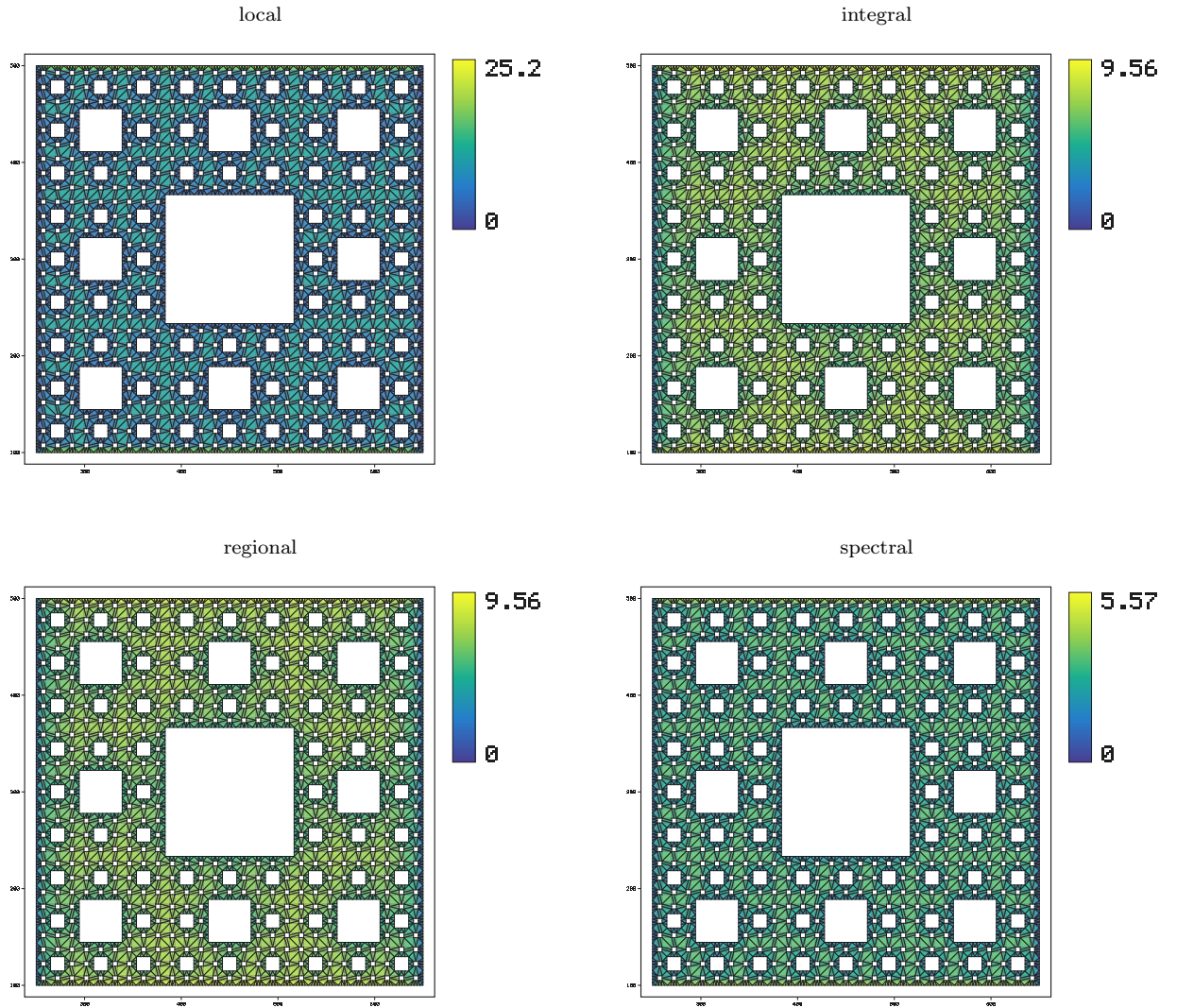


Figure 4.3: Two-dimensional solution fields for the Sierpinski carpet type domain at $s = 0.5$.

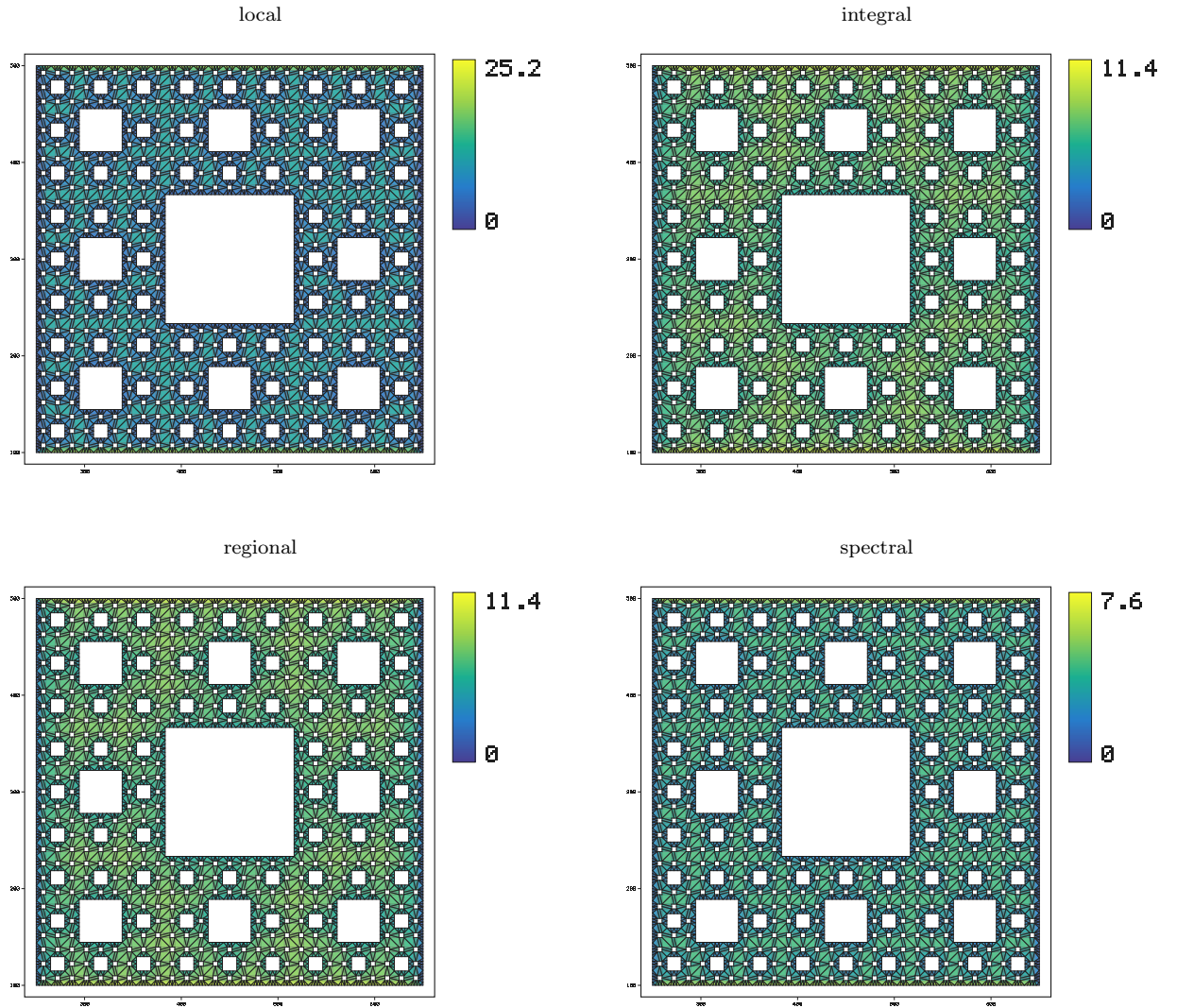


Figure 4.4: Two-dimensional solution fields for the Sierpinski carpet type domain at $s = 0.6$.

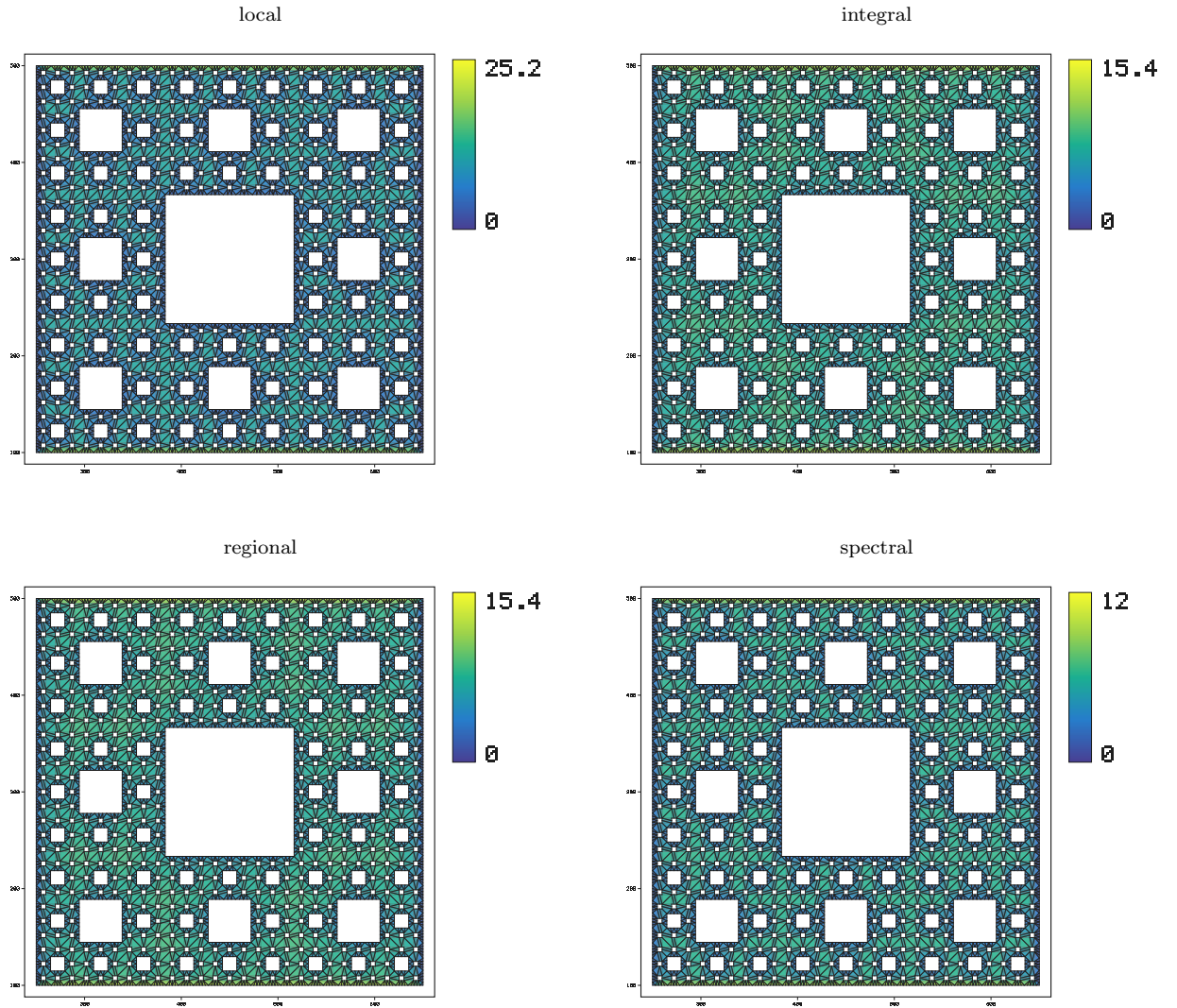


Figure 4.5: Two-dimensional solution fields for the Sierpinski carpet type domain at $s = 0.75$.

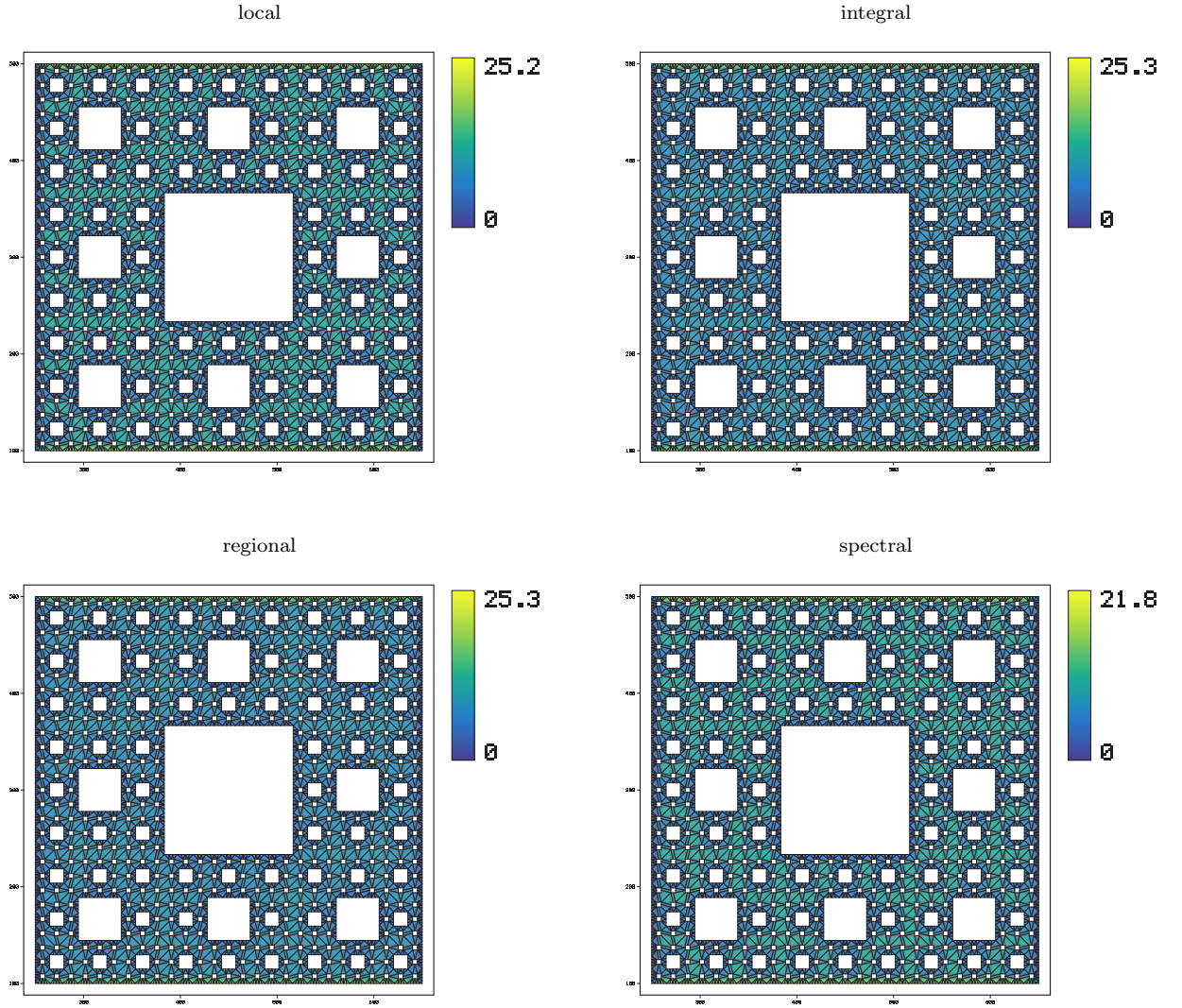


Figure 4.6: Two-dimensional solution fields for the Sierpinski carpet type domain at $s = 0.95$.

other. The spectral realization, on the contrary, is closer in color scale to the local one, but at the same time it looks more even: the local amplifications near the complex pore boundary are weaker. This agrees with the previous quantitative results, where exactly the spectral model yielded the smallest flux through the pore boundary.

For $s = 0.6$, in Figure 4.4, the tendency is preserved, but the contrast between the realizations is already smaller. The integral and regional formulations still practically coincide with each other, whereas the spectral realization remains visually smoother. At the same time, all four fields are already closer to one another than at $s = 0.5$, which corresponds to the decrease in the differences in normalized fluxes and energies.

In Figure 4.5, for $s = 0.75$, the discrepancies become even less noticeable. The local, integral, and regional realizations have very close locations of the zones of higher and lower values, and the spectral model, although it differs, is no longer as sharply distinct as for smaller s . Thus, as the fractional order increases, the visual picture gradually approaches the one obtained for the local model.

Finally, in Figure 4.6, at $s = 0.95$, the integral and regional realizations practically do not differ from the local one not only in numerical characteristics but also in the appearance of the fields themselves. The spectral realization also becomes closer to them, but a certain additional smoothing of the field is still preserved. Thus, the new visualization confirms the general conclusion of the numerical experiments: the integral and regional models in this geometry are very close to the local one, whereas the spectral realization produces a somewhat different picture of the distribution of values, especially for smaller values of s .

The three-dimensional surfaces in Figure 4.7 are convenient because they show not only the location of the zones of maximum and minimum, but also the intensity of the change of the solution. In the local model, the surface has the most pronounced relief: the rises and drops are sharper, and therefore the

Local model

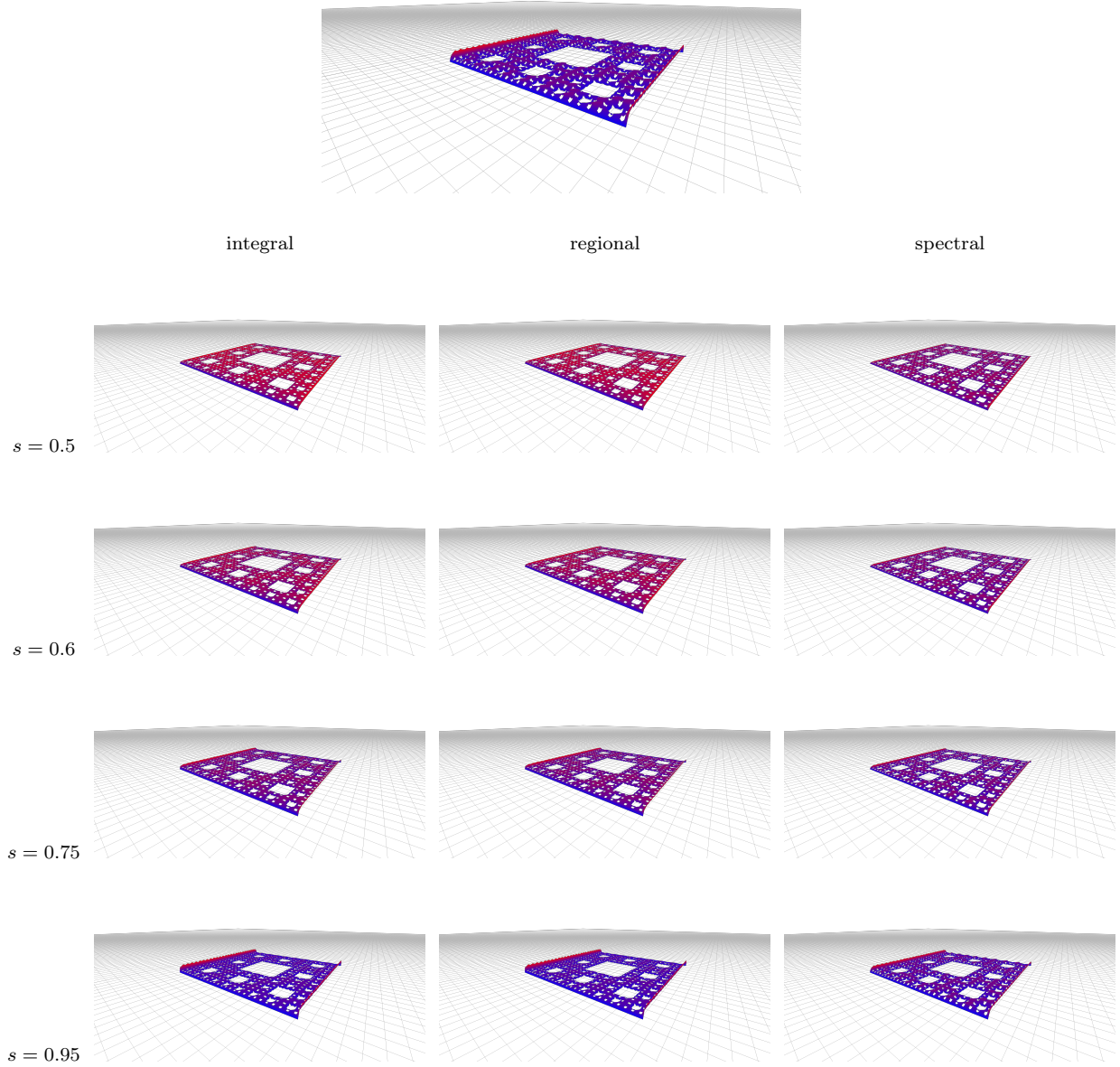


Figure 4.7: Three-dimensional solution visualizations for the Sierpinski carpet type domain for the values $s = 0.5$, $s = 0.6$, $s = 0.75$, and $s = 0.95$.

gradient structure in the domain is the strongest. For the integral and regional realizations, this relief is partially smoothed, especially for smaller s , but it preserves the overall form of the local distribution. This once again confirms that these two formulations modify the local transport mechanism quantitatively, but do not change its qualitative spatial picture.

For the spectral realization, the surface is the flattest among all the compared models, especially at $s = 0.5$ and $s = 0.6$. As s increases, the amplitude grows, but even at $s = 0.95$ the shape of the surface remains visually smoother than in the local case. Taken together with the two-dimensional maps and the integral characteristics, this gives a consistent picture: the integral and regional realizations are close to the local one not only numerically, but also morphologically, whereas the spectral realization forms a different model of spatial redistribution, more sensitive to the fractional order and less prone to local concentration of flux near the complex internal boundary.

Conclusions

In this work, a numerical approach to modeling diffusion in domains with a complex internal structure for local and spatially nonlocal operators of fractional type has been constructed and investigated. Within the framework of the work, local, integral, regional, and spectral formulations of the problem have been stated, their finite element approximations have been constructed, and software has been developed for geometry generation, mesh construction, assembly of discrete operators, carrying out computational experiments, and visualization of the results.

The numerical results show that for perforated and nanoporous domains the local model does not in all cases fully reflect the features of the diffusion process. In media with a developed internal surface, a large number of pores, and narrow channels, spatially nonlocal interaction may acquire essential significance. In this sense, fractional operators can be regarded as a meaningful generalization of the classical model, which makes it possible to take into account long-range interaction and the sensitivity of the solution to complex internal geometry. The comparison carried out has shown that, within the framework of this study, both the value of the fractality parameter and the type of realization of the fractional operator noticeably influence the solution field, the intensity of exchange through the internal boundaries, and the integral characteristics of the process.

The significance of the obtained results lies in the fact that the constructed models and software tools can be used for investigating transport processes in porous and multiscale media, where classical local equations do not provide a

sufficiently accurate description. At the same time, the developed approach has a broader character and can be applied in problems where complex geometry substantially affects the quality of mathematical modeling and numerical reconstruction.

It is worth noting the connection of this work with problems of computer vision and spatial reconstruction. The approaches to geometric modeling, mesh representation of complex surfaces, and variational data processing constructed in this work are consistent with the line of research presented in [12], where a variational-computational approach to monocular 3-D reconstruction of the face surface is proposed. Within the framework of this work, such a connection is of a rather theoretical character: it shows the possibility of combining the approaches constructed here with the results of [12] and of their further generalization. This opens the prospect of future studies in which methods of numerical analysis, finite element discretization, and geometric data processing can be used for constructing more accurate and stable approaches to reconstructing complex surfaces from visual information.

Separately, let us note that the extended comparison of the integral, regional, and spectral realizations of spatially nonlocal diffusion in perforated domains also constitutes the basis for the article submitted for publication [11], devoted to the comparative finite element analysis of these formulations.

Further development of the work should be directed toward the generalization of the constructed models to three-dimensional domains, toward full nonstationary fractional problems, and also toward combining spatially nonlocal models with 3-D reconstruction problems, in particular under conditions of incomplete data, occlusions, and complex surface geometry.

At the same time, the work also has a number of limitations that are important to record clearly. The study has been carried out for a limited class of geometries and parameters, and the main numerical conclusions have been obtained on the basis of two-dimensional experiments. The developed software

already supports the nonstationary local problem and demonstrates the time evolution of the solution and its convergence to a stationary state; however, for fractional realizations of the operator it was not possible to perform a full series of nonstationary computations, since such calculations proved to be computationally too demanding. In addition, within the framework of this work, a complete study of convergence and efficiency has not been carried out for all spatially nonlocal formulations. Therefore, the shortcomings of the work include the limited scope of the computational experiment, the absence of a full nonstationary analysis for the fractional models, and the need for further extension of both the mathematical and the software parts of the study.

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Appendix

Declaration on the use of artificial intelligence tools.

During the preparation of this work, artificial intelligence tools were used as an auxiliary means for language editing, clarification of formulations, technical formatting of individual fragments of the text, and checking the consistency of the presentation. The formulation of problems, mathematical models, numerical schemes, software implementation, computational experiments, and interpretation of results were carried out by the author independently.

QR code to the source code.



<https://github.com/anshelino0606/DelaunayForge>